Session 4.1: Variant Effect Predictors I – Introduction to VEPs

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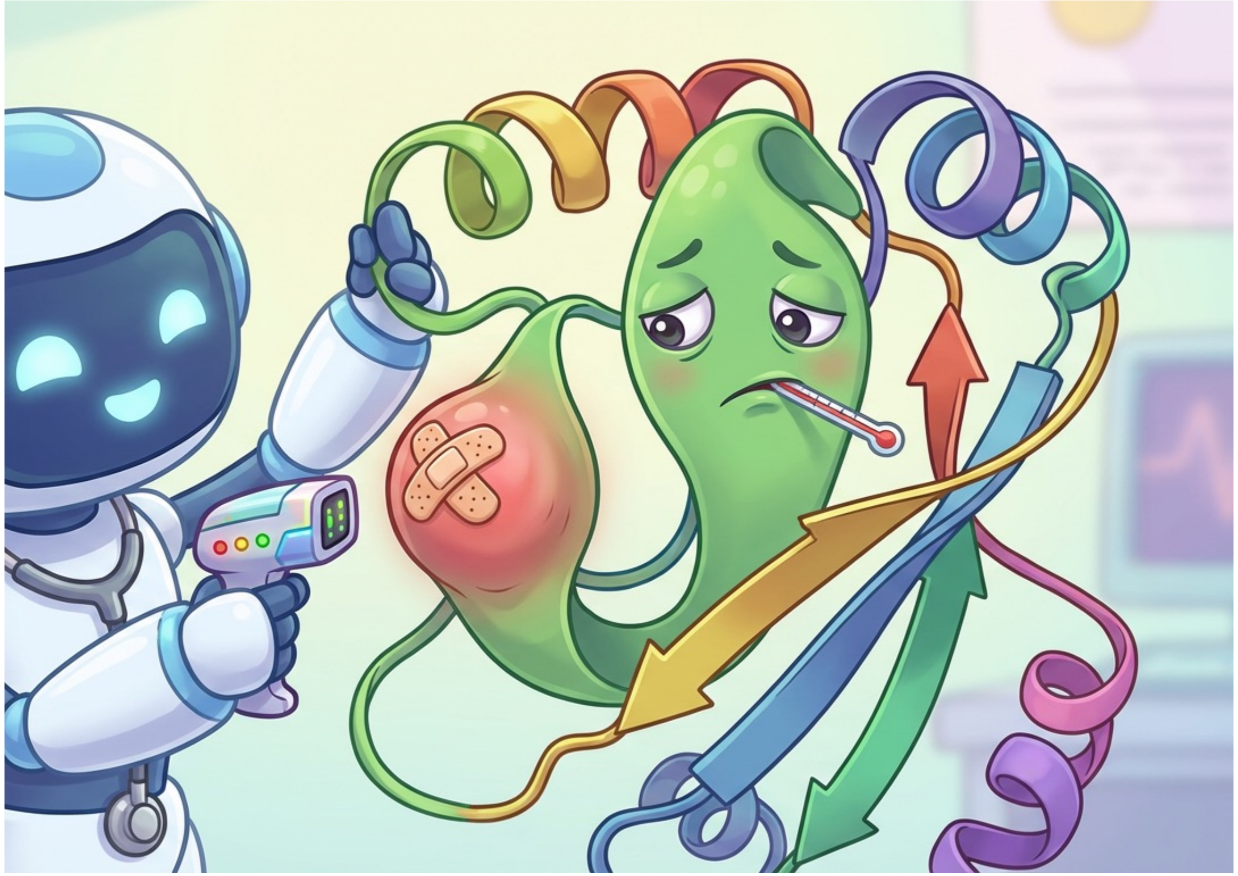
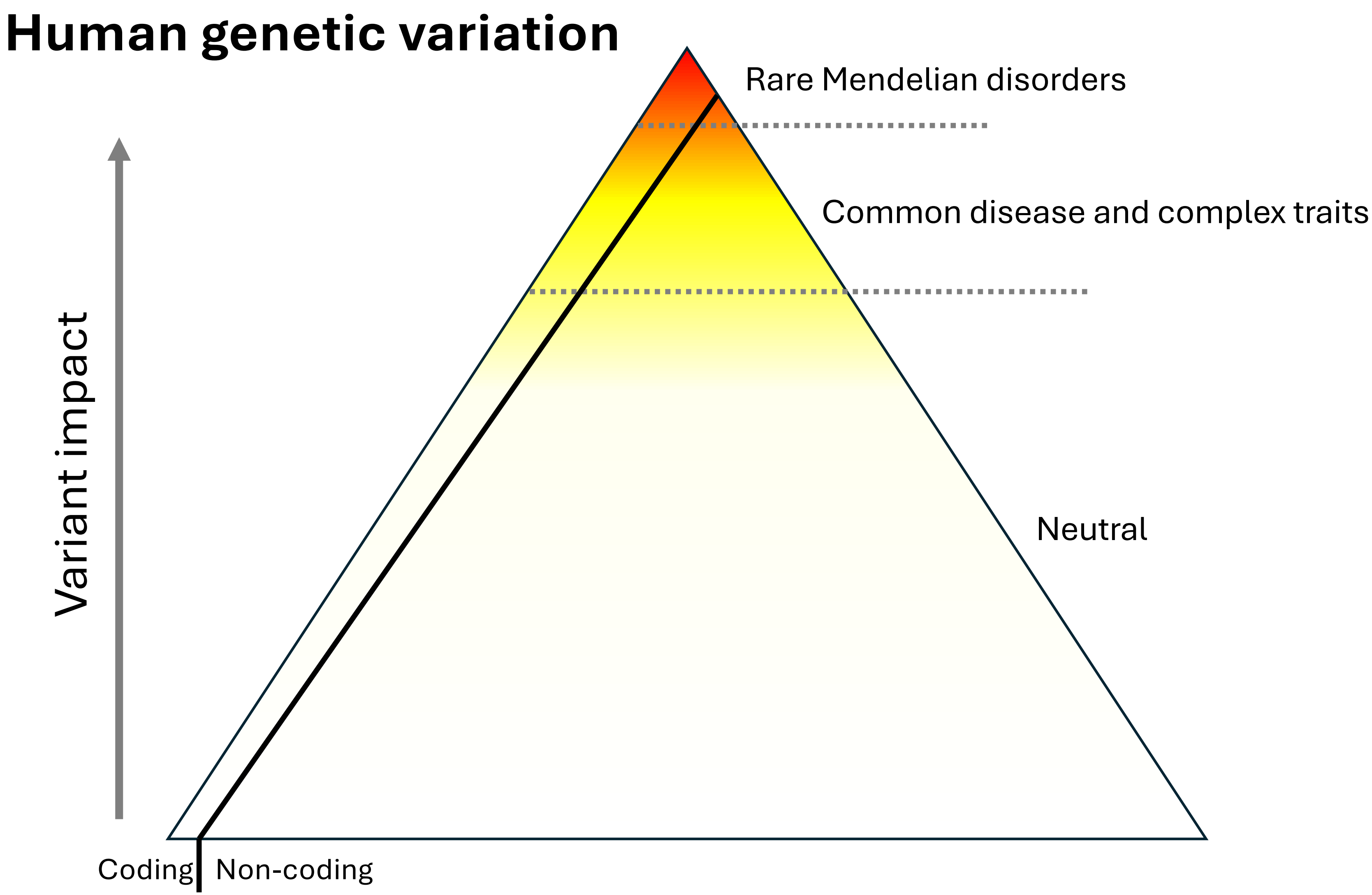
Links and datasets

https://sites.google.com/view/mave2025-lectures

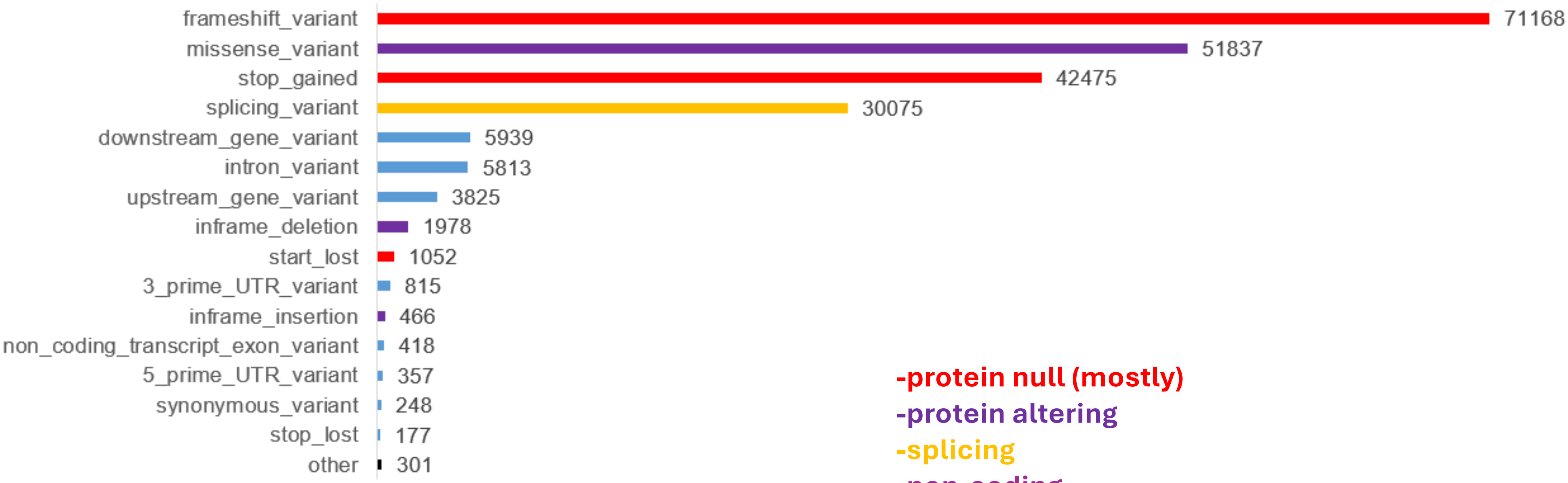
Variant effect predictors (VEPs)

- VEPs are like in silico MAVEs
- Different types:
  - fitness, pathogenicity, stability, binding, splicing, etc
  - missense, indels, synonymous, non-coding
- Lots of names, but field is converging on VEP
  - Different from Ensembl VEP!
- Here, we are mostly interested in predicting disease and improving diagnosis for *missense* variants




Human genetic variation
Rare Mendelian disorders
Common disease and complex traits
Neutral
Variant impact
Coding
Non-coding

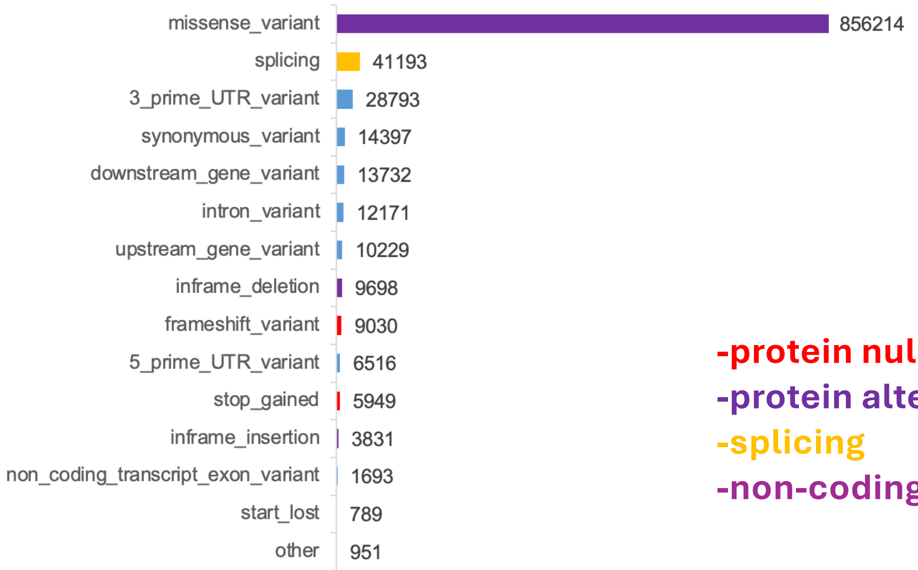
Types of pathogenic mutations



-protein null (mostly)
-protein altering
-splicing
-non-coding

ClinVar
Clinically relevant variation

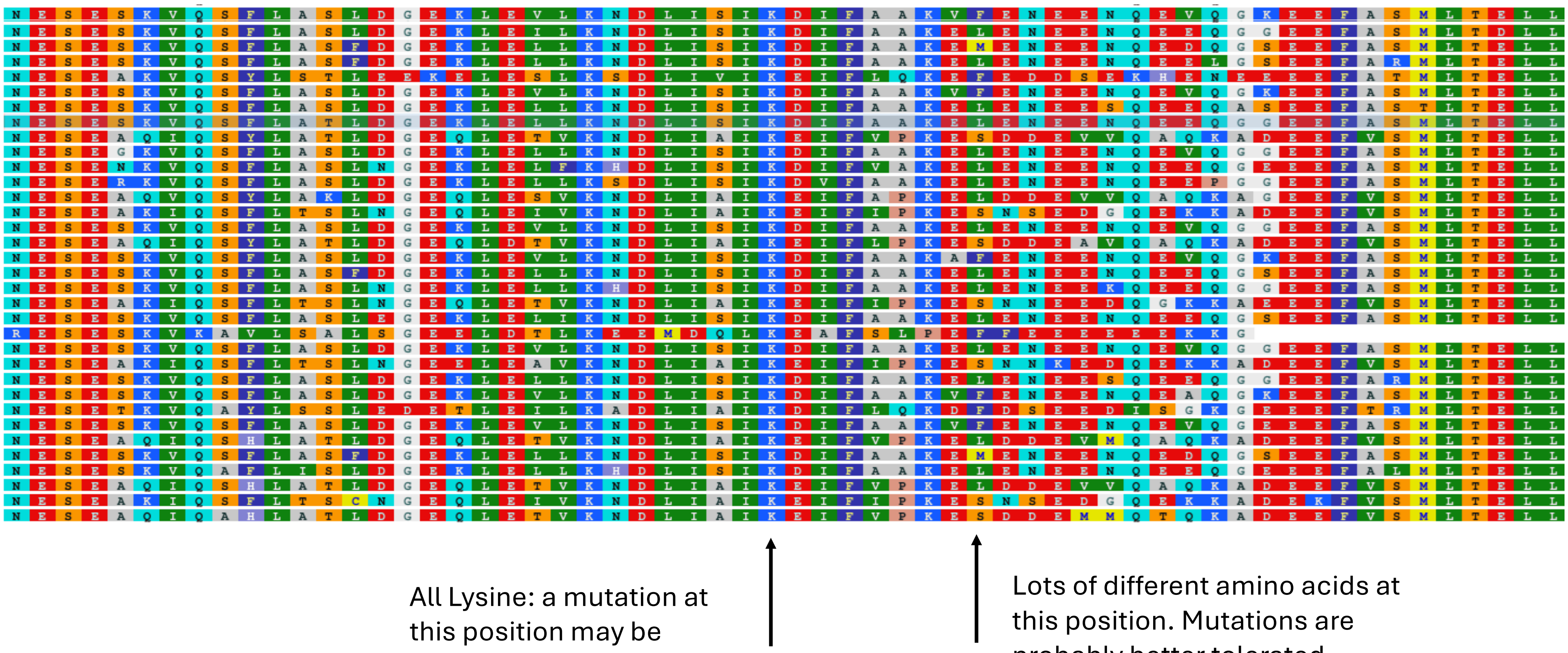
Variants of uncertain significance (VUS) - *variants that have been specifically evaluated for their potential to cause disease, but there is currently not enough evidence to classify them as either benign or pathogenic*



-protein null (mostly)
-protein altering
-splicing
-non-coding/synonymous

ClinVar
Clinically relevant variation

Nearly all variant effect predictors make use of evolutionary conservation

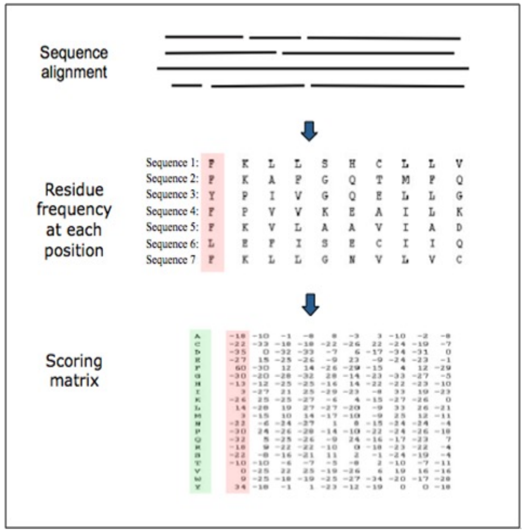


The Foundation of VEPs

- **Substitution matrices (1970s)**
  - Observed frequencies of amino-acid changes in related proteins
  - Global scores (same for every position)
  - BLOSUM62 still surprisingly competitive today

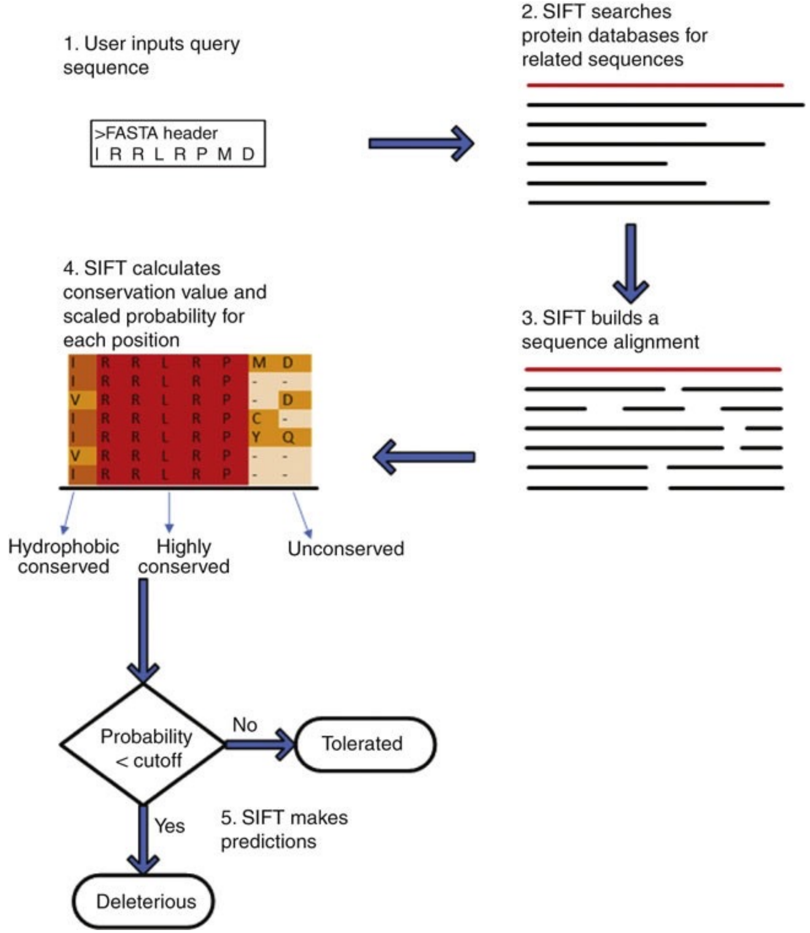
- **Position-specific scoring (late 1990s)**
  - Instead of one global matrix → calculate tolerated substitutions per alignment column
  - PSIC profiles (Sunyaev et al. 1999) → First big accuracy leap because conservation is now context-aware

	C	S	T	A	G	P	D	E	Q	N	H	R	K	M	I	L	V	W	Y	F	
C	9																				C
S	-1	4																			S
T	-1	1	5																		T
A	0	1	0	4																	A
G	-3	0	-2	0	6																G
P	-3	-1	-1	-1	-2	7															P
D	-3	0	-1	-2	-1	-1	6														D
E	-4	0	-1	-1	-2	-1	2	5													E
Q	-3	0	-1	-1	-2	-1	0	2	5												Q
N	-3	1	0	-2	0	-2	1	0	0	6											N
H	-3	-1	-2	-2	-2	-2	-1	0	0	1	8										H
R	-3	-1	-1	-1	-2	-2	-2	0	1	0	0	5									R
K	-3	0	-1	-1	-2	-1	-1	1	1	0	-1	2	5								K
M	-1	-1	-1	-1	-3	-2	-3	-2	0	-2	-2	-1	-1	5							M
I	-1	-2	-1	-1	-4	-3	-3	-3	-3	-3	-3	-3	-3	1	4						I
L	-1	-2	-1	-1	-4	-3	-4	-3	-2	-3	-3	-2	-2	2	2	4					L
V	-1	-2	0	0	-3	-2	-3	-2	-2	-3	-3	-3	-2	1	3	1	4				V
W	-2	-3	-2	-3	-2	-4	-4	-3	-2	-4	-2	-3	-3	-1	-3	-2	-3	11			W
Y	-2	-2	-2	-2	-3	-3	-3	-2	-1	-2	2	-2	-2	-1	-1	-1	-1	2	7		Y
F	-2	-2	-2	-2	-3	-4	-3	-3	-3	-3	-1	-3	-3	0	0	0	-1	1	3	6	F
	C	S	T	A	G	P	D	E	Q	N	H	R	K	M	I	L	V	W	Y	F	



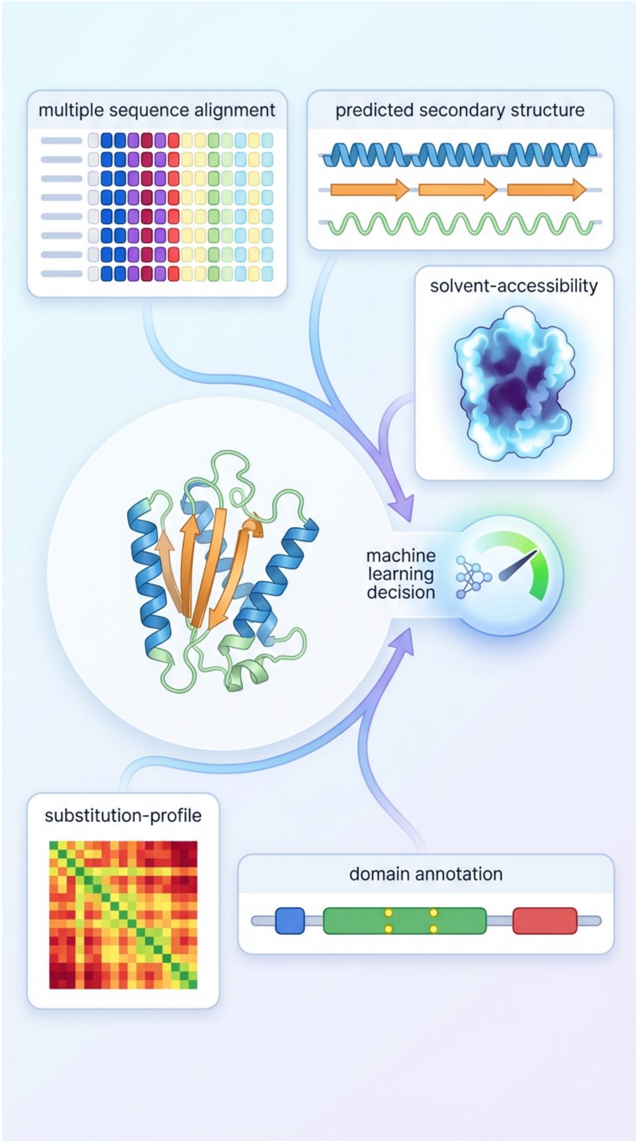
SIFT (2001) – the first widely used VEP

- Sorting Intolerant From Tolerant (SIFT)
- Builds a protein-specific multiple sequence alignment (MSA)
- Adds Dirichlet pseudocounts for unobserved residues
- Simple, fast, interpretable → still a common baseline in 2025



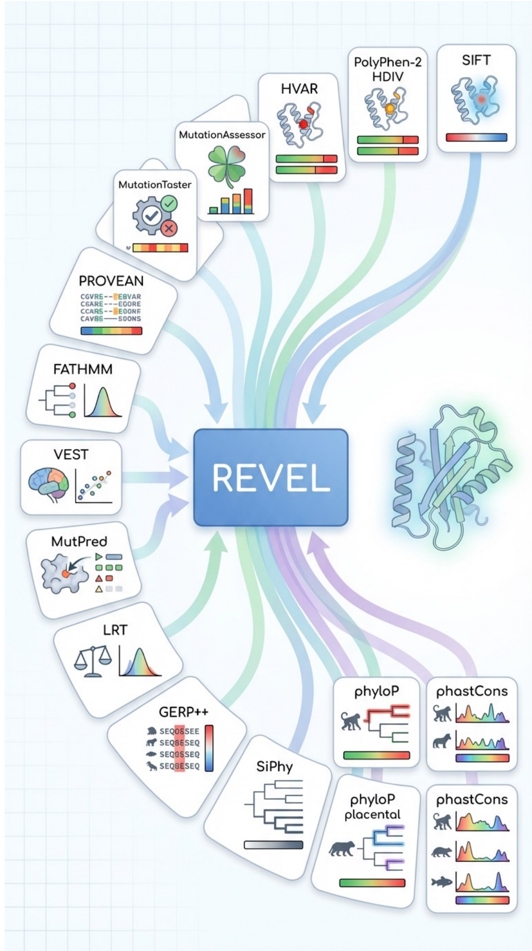
PolyPhen-2: early multi-feature prediction with machine learning

- **PolyPhen (2002)**
  - Extended beyond conservation.
  - Added structural context (known structures when available).
  - Incorporated protein annotations: domains, active sites, ligand-binding residues, catalytic residues.
  - Used simple rules and a decision tree.
- **PolyPhen-2 (2010)**
  - A major shift toward supervised machine learning (naïve Bayes classifier), **trained to predict pathogenicity**.
  - Integrated an even broader set of features:
    - Secondary structure predictions.
    - Solvent accessibility.
    - Known functional sites.
    - Sequence features around the variant.
  - Became the most widely used "early generation" machine learning VEP, still very widely used today



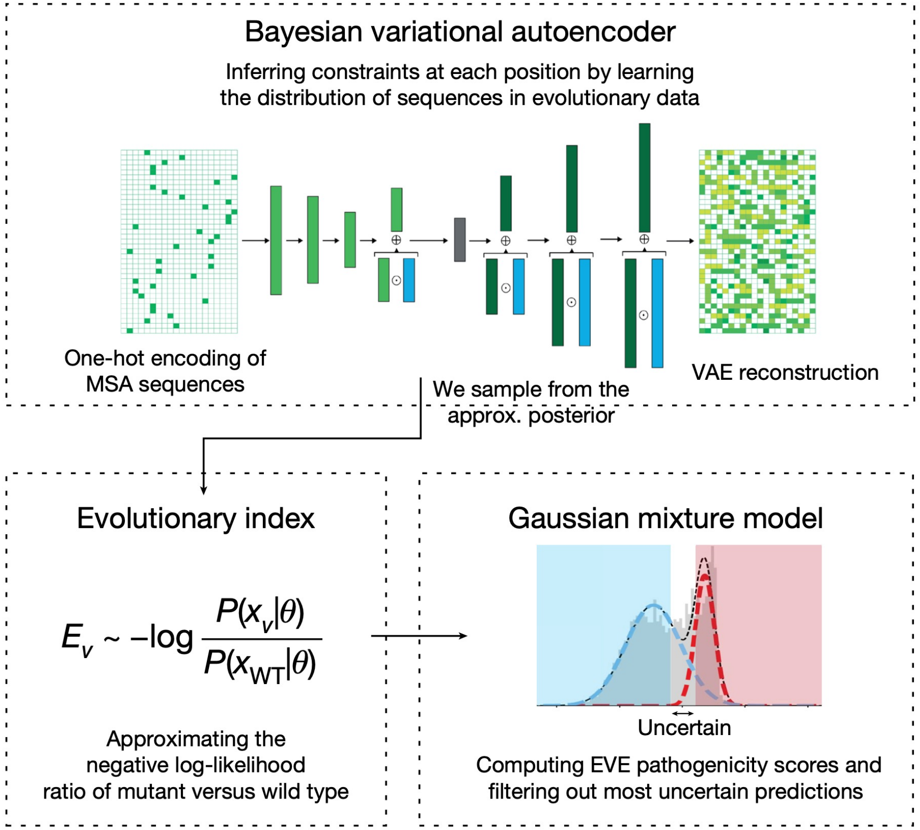
Meta-predictors: combining many VEPs into a single consensus score

- As the number of individual VEPs grew, many produced conflicting outputs for the same variant.
- Meta-predictors emerged to combine the strengths of multiple VEPs rather than relying on any single method.
- These tools take the outputs of many existing predictors as input features, and use supervised learning to produce a more stable overall score.
- They became widely used because their consensus approach generally improves robustness across proteins and variant types.
- **REVEL (2016)**
  - One of the most influential meta-predictors.
  - Combines outputs from many VEPs, plus conservation and other sequence features.
  - Uses a random forest to integrate these heterogeneous scores into a final probability-like output.
  - Became a de facto standard in clinical variant interpretation pipelines.



EVE (2021): deep generative modelling of evolutionary sequences

- EVE is **unsupervised**: it uses **no clinical labels** at any stage of training.
- Uses only **evolutionary sequence alignments** (~250 million UniRef sequences searched to build per-protein MSAs).
- Trains a Bayesian variational autoencoder (VAE) to learn the joint distribution of amino acids across the alignment. This captures:
  - residue conservation
  - patterns of co-variation
  - higher-order dependencies
- Produces an **evolutionary index** for each variant (approximate log-likelihood ratio of variant vs wild type).
- A simple **Gaussian mixture model** converts these indices into benign vs pathogenic probabilities, giving the **EVE score** (0–1).



AlphaMissense (2023): large-scale protein language and structure modelling

- **Foundation:** AlphaMissense builds on AlphaFold's architecture, using a **protein language model** and **structural representations** tailored for single amino acid variants.
- **Training data:**
  - Trained on **biological sequences**metagenomes and related resources. , including UniRef, MGnify
  - Importantly, **no supervised clinical** were used **However, the output calibration is tuned using population allele frequency data** from gnomAD, which influences the pathogenicity thresholding and score interpretation.
- **Model design:**
  - Modified AlphaFold-like architecture with masked-token objectives and structural context modules.
  - Output is a continuous pathogenicity probability based on variant-specific features.
- **Scale:**
  - Scores all 216 million possible human missense variants.
  - Released as a genome-wide resource.
  - Be careful of the labels 'likely pathogenic' and 'likely benign'

Wednesday 20 September 2023 The Guardian

National

AI tool could help speed up diagnosis of rare disorders

Ian Sample
Science editor

Scientists at Google DeepMind have built an artificial intelligence program that can predict whether millions of genetic mutations are either harmless or likely to cause disease, in an effort to speed up research and the diagnosis of rare disorders.

The program makes predictions about so-called missense mutations, where a single letter is misspelt in the DNA code. Such mutations are often harmless but they can disrupt how proteins work and cause diseases from cystic fibrosis and sickle-cell anaemia to cancer and problems with brain development.

The researchers used AlphaMissense to assess all 71m single-letter mutations that could affect human proteins. When they set the program's precision to 90%, it predicted that 57% of missense mutations were probably harmless and 32% were probably harmful. It was uncertain about the impact of the rest.

A free online catalogue of the predictions has been released to help geneticists and clinicians studying how mutations drive diseases or diagnosing patients with rare disorders.

A typical person has about 9,000 missense mutations throughout their genome. Of more than 4m seen in humans, only 2% have been classified as either benign or pathogenic. Doctors already have computer programs to predict which mutations may drive disease but as the predictions are inaccurate, they can only provide supporting evidence for a diagnosis.

Writing in Science, Dr Jun Cheng and others describe how AlphaMissense performs better than current "variant effect predictor" programs and should help experts pinpoint more swiftly which mutations are driving diseases. The program may also flag mutations that have not previously been linked to specific disorders and guide doctors to better treatments.

AlphaMissense was fed DNA data from humans and closely related primates to learn which mutations are common, and therefore probably benign, and which are rare and potentially harmful.

The program also familiarised itself with the "language" of proteins by studying millions of protein sequences and learning what a "healthy" protein looks like. When the trained AI is fed a mutation, it generates a score to reflect how risky the genetic change appears to be.

"This is very similar to human language," Cheng said. "If we substitute a word in an English sentence, a person familiar with English can immediately see whether the word substitution will change the meaning of the sentence or not."

Prof Joe Marsh, a computational biologist at Edinburgh University who was not involved in the work, said AlphaMissense had "great potential".

"We have this issue with computational predictors where everybody says their new method is the best," he said. "You can't really trust people, but [the DeepMind researchers] do seem to have done a pretty good job."

If clinical experts decided that AlphaMissense was reliable, its predictions may carry more weight in future disease diagnosis, he said.

THE WALL STREET JOURNAL.

SCIENCE

Google DeepMind's AI Model Scours Our Genes to Guess Who Might Get Sick

A machine-learning model evaluated 71 million variations in human proteins for their likelihood to cause disease

In a related article in Science, Joseph A. Marsh, chair of computational protein biology at the University of Edinburgh, and Sarah A. Teichmann, head of cellular genetics at the Wellcome Sanger Institute, who weren't involved with the project, applauded the work but said its current utility is minimal.

"Current computational predictors are not considered reliable enough to be used by themselves for genetic diagnosis," Marsh said.

NEWS | 19 September 2023 Nature News

AlphaFold tool pinpoints protein mutations that cause disease

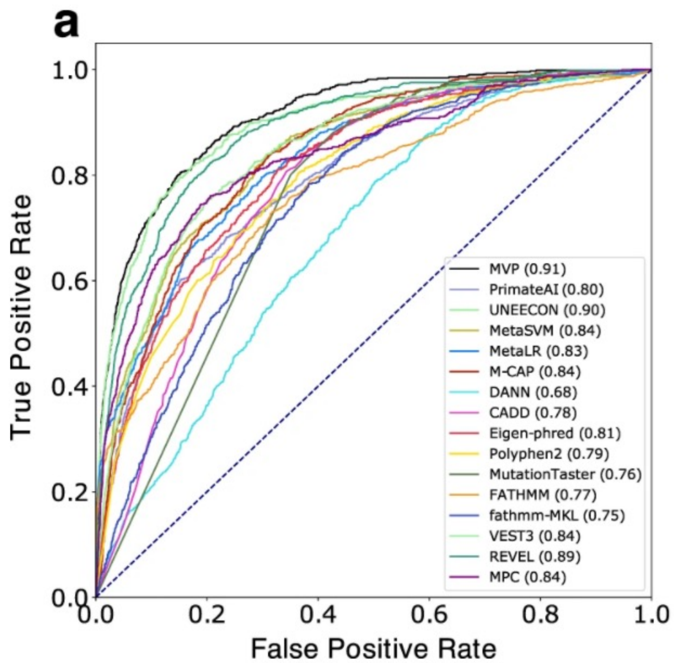
Researchers have adapted the AI network to search for genetic changes linked to ill health.

Its impact won't be as significant as AlphaFold, which ushered in a new era in computational biology, agrees Joseph Marsh, a computational biologist at the MRC Human Genetics Unit in Edinburgh, UK. "It's exciting. It's probably the best predictor we have right now. But will it be the best predictor in two or three years? There's a good chance it won't be."

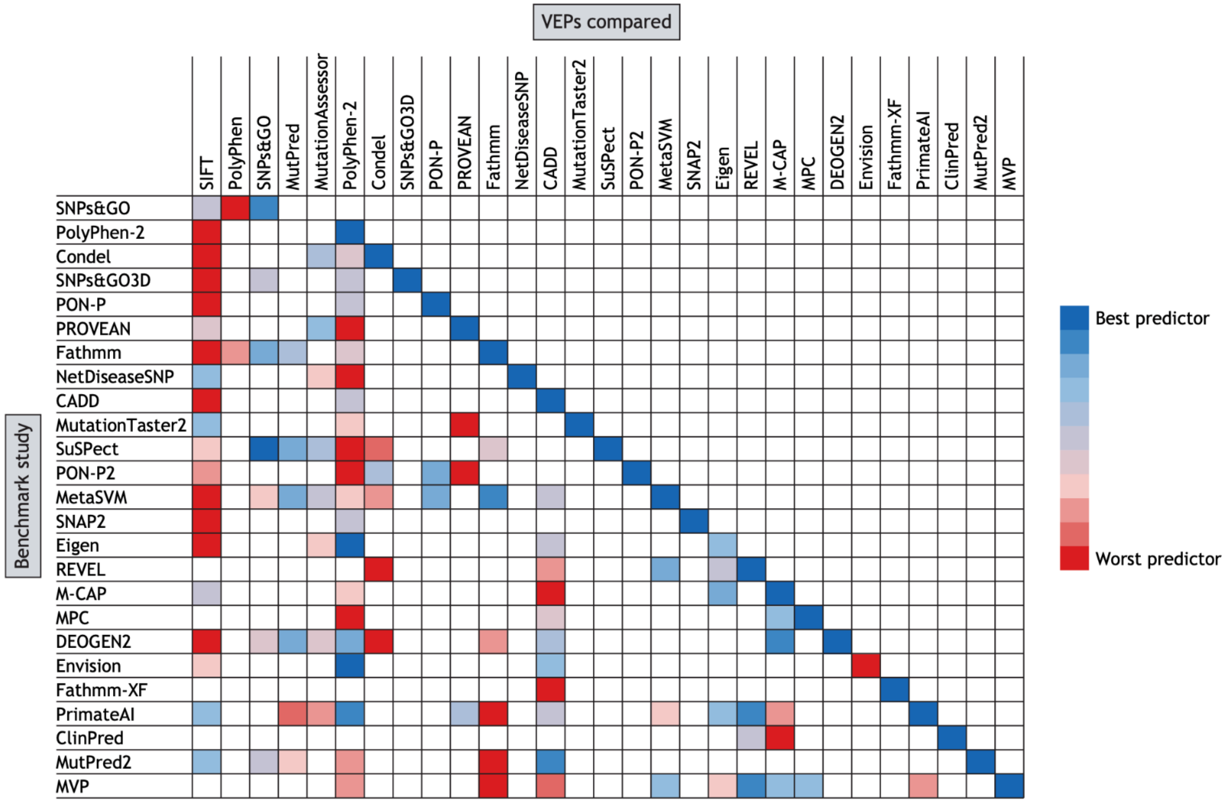
Computational predictions currently have a minimal role in diagnosing genetic diseases, says Marsh, and recommendations from physicians' groups say that these tools should provide only supporting evidence in linking a mutation to a disease. AlphaMissense confidently classified a much larger proportion of missense mutations than have previous methods, says Avsec. "As these models get better than I think people will be more inclined to trust them."

How VEP performance is usually assessed

- With so many predictors claiming improvements, we need a way to compare them.
- The field almost always uses the same evaluation setup: classifying missense variants as pathogenic or benign.
- Pathogenic variants
  - Usually from ClinVar (Pathogenic / Likely Pathogenic).
- Benign variants
  - ClinVar Benign / Likely Benign, or
  - ’Putatively benign’ population variants from gnomAD.
- How performance is measured
  - Predictors output continuous scores, but evaluation is framed as a binary classification task.
  - A receiver operating characteristic (ROC) curve shows the relationship between true positive rate and false positive rate across all score thresholds.
  - AUC summarises the ROC: AUC is the probability that a randomly chosen pathogenic variant is scored higher than a randomly chosen benign one.
- **Important: leave out any variants used in training!**



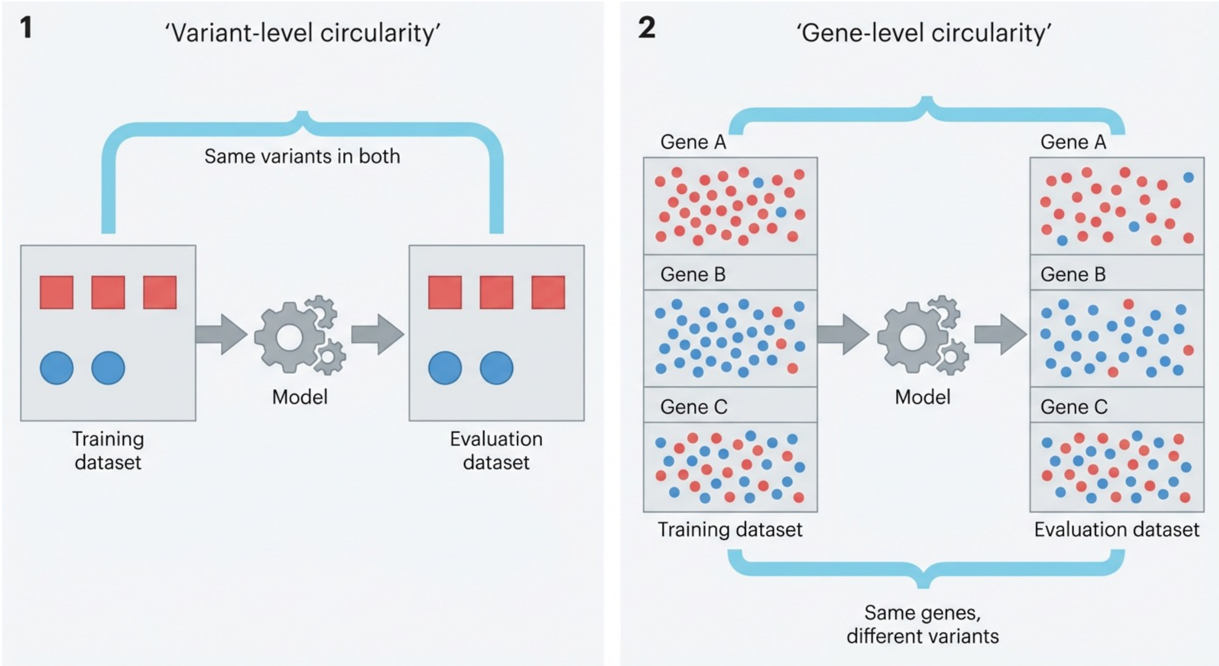
(Almost) every VEP claims to be the best



Livesey & Marsh (2022) Dis Mod Mech

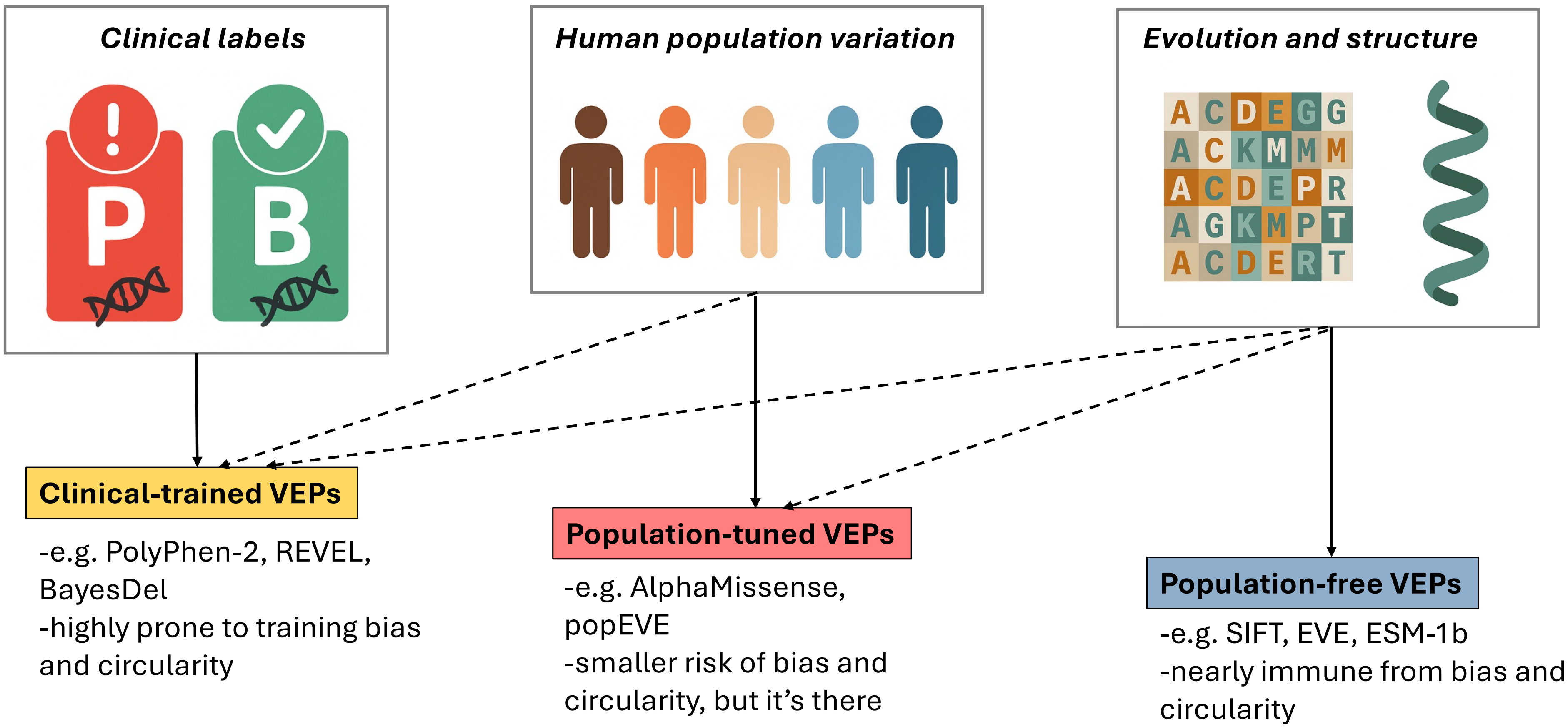
The problem of training bias and data circularity

- ***Variant-level (type 1) circularity*** – VEPs learn to classify specific variants during training
- ***Gene-level (type 2) circularity*** – VEPs learn associations between certain genes and pathogenicity



These circularities are very difficult to get rid of...

Classifying VEPs based on the information they use



How can we compare VEPs in a fair manner?

- Our strategy: look at correlations across lots of deep mutational scanning (i.e. MAVE) datasets
  - Livesey & Marsh (2020) Mol Syst Biol
  - Livesey & Marsh (2023) Mol Syst Biol
  - Livesey & Marsh (2025) Genome Biol

Ben Livesey

Livesey and Marsh *Genome Biology* (2025) 26:104
https://doi.org/10.1186/s13059-025-03575-w

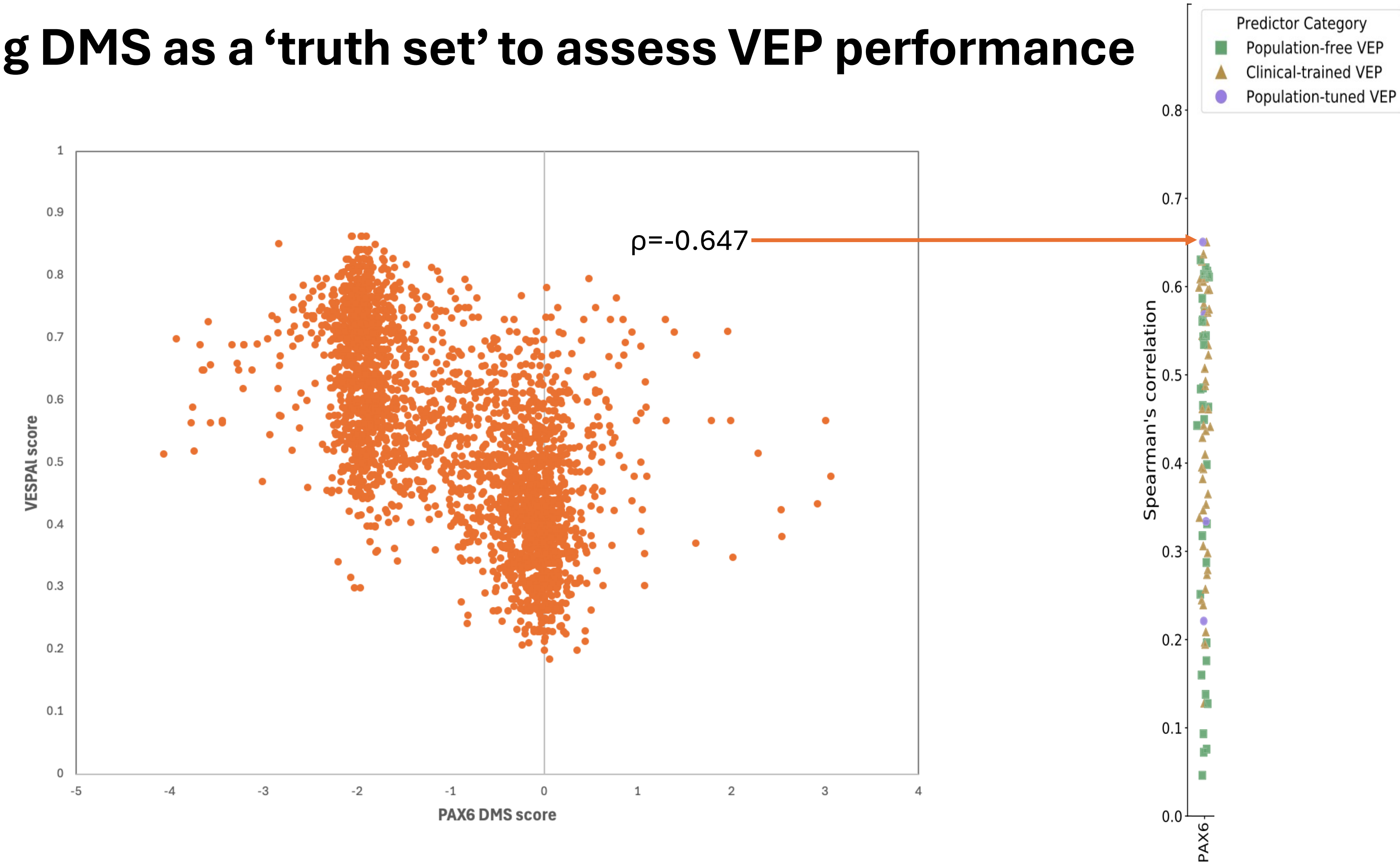
Genome Biology

RESEARCH Open Access

Variant effect predictor correlation with functional assays is reflective of clinical classification performance

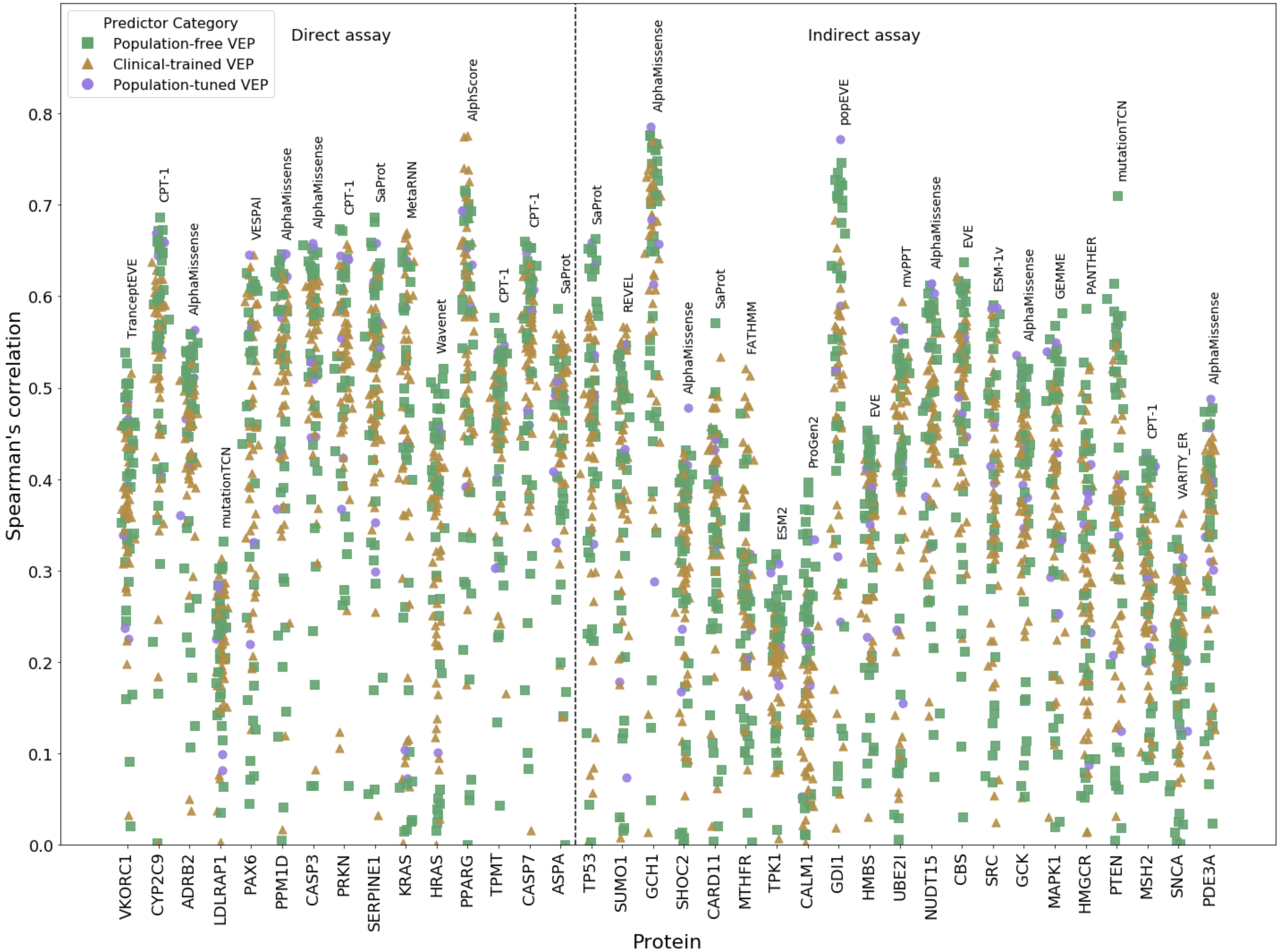
Benjamin J. Livesey[1] and Joseph A. Marsh[1*]

Using DMS as a ‘truth set’ to assess VEP performance



VEP and MAVE agreement

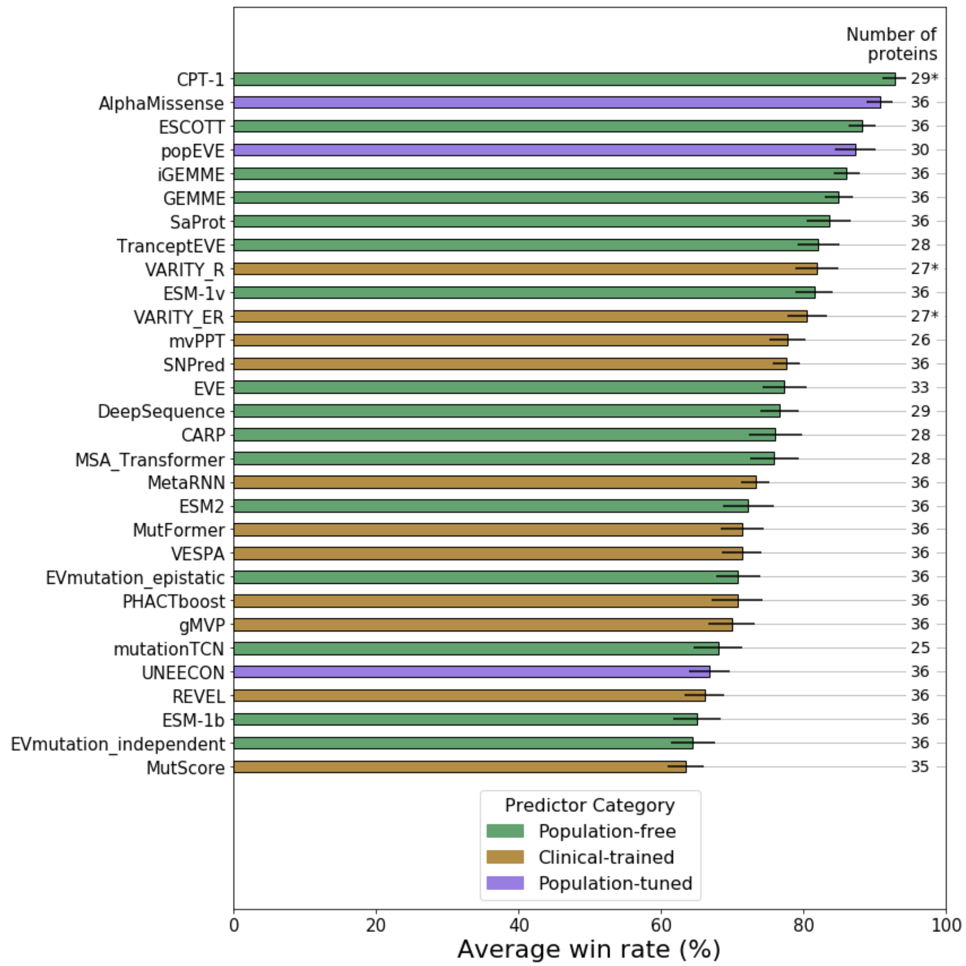
- Spearman's correlation between 96 VEPs and 36 MAVE datasets.
- Average correlation ~0.6 for top predictors.
- Agreement varies a lot depending on the protein/MAVE assay.
- Some VEPs are consistently among the top performers.



Pairwise competition strategy

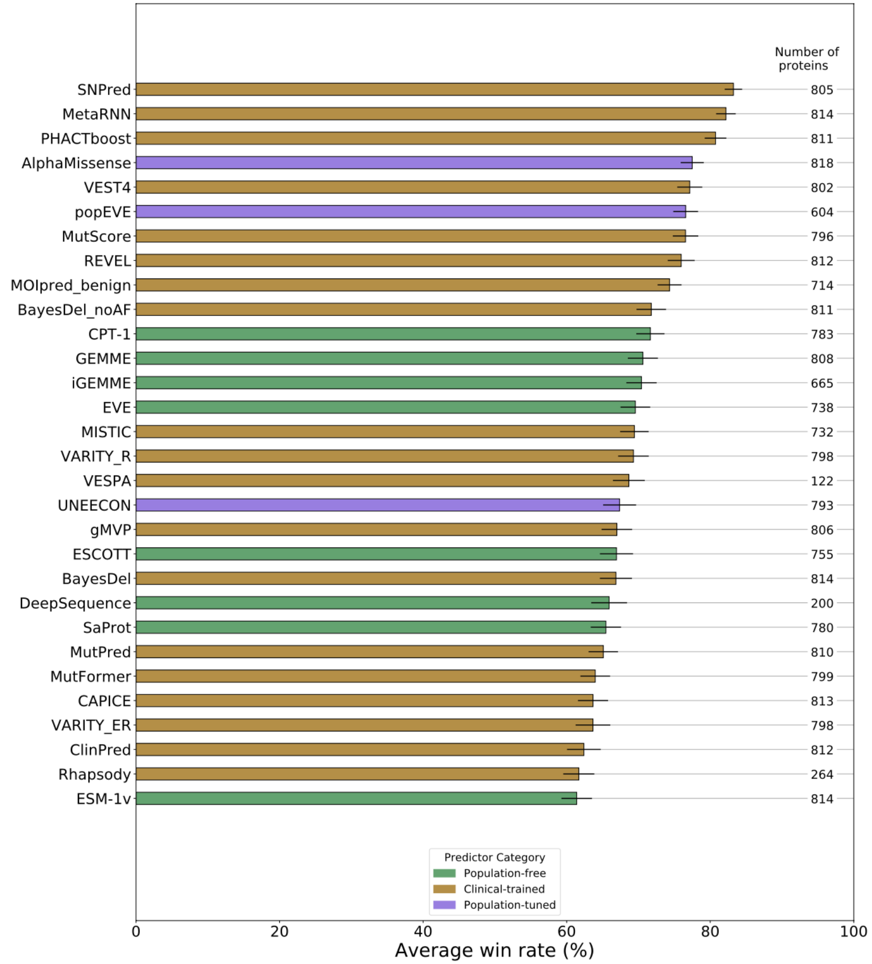
- For each pair of VEPs, select only the variants shared between both VEPs and present in the DMS
- Whichever has the highest Spearman’s correlation “wins”
- Calculate the win rate across all DMS datasets
- Calculate the average win rate for each VEP against all others

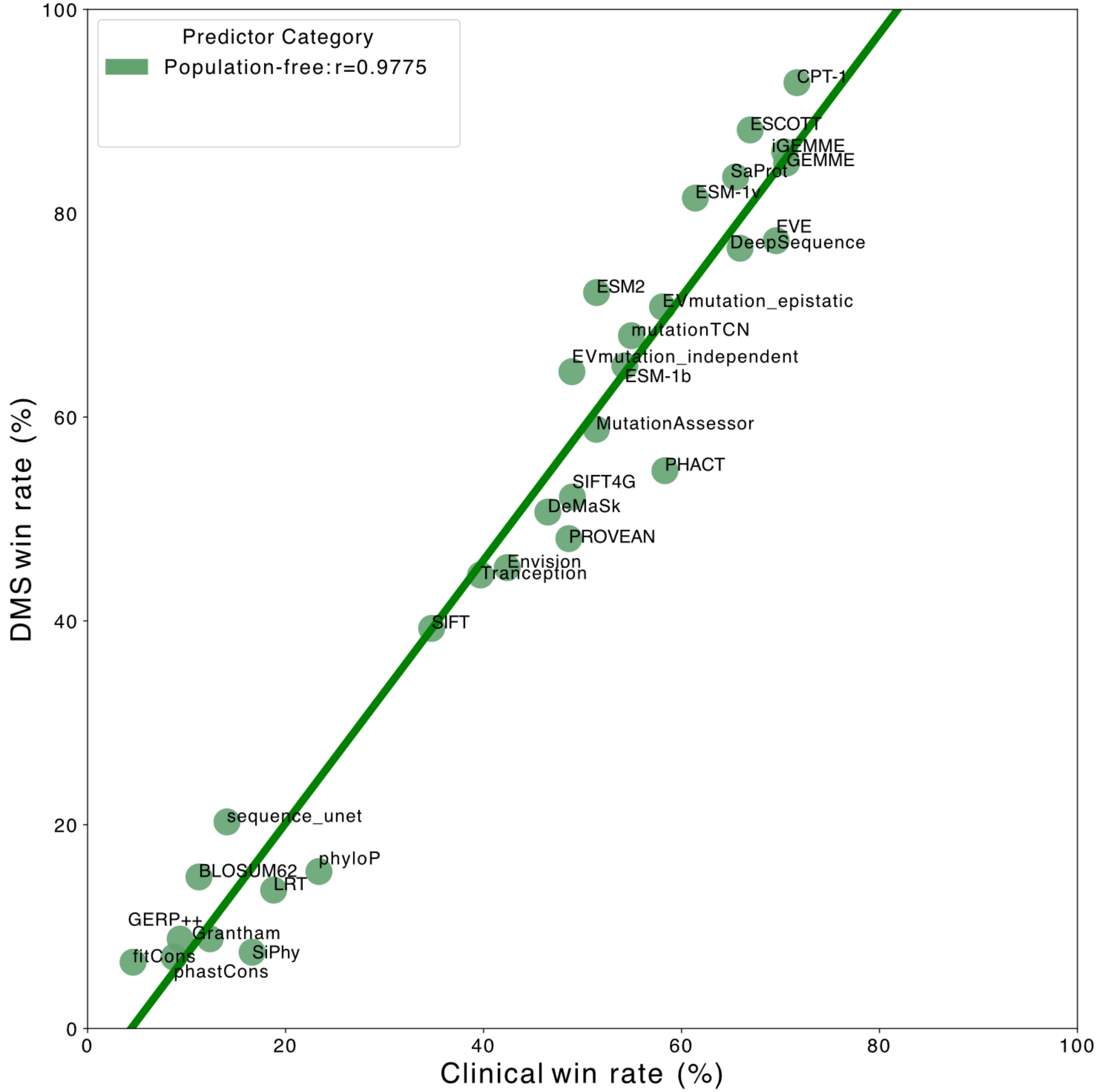
Top 30 VEPS out of 97 tested



Do these rankings reflect pathogenicity prediction?

- We can apply the same pairwise competition strategy to 1034 disease genes with at least 10 pathogenic and 10 putatively benign missense variants
- AUROC (or AUPRC) instead of Spearman's
- The use of rare gnomAD for putatively benign reduces circularity





Predictor Category
Population-free: r=0.9775
CPT-1
ESCOTT
iGEMME
GEMME
SaProt
ESM-1v
EVE
DeepSequence
ESM2
EVmutation_epistatic
mutationTCN
EVmutation_independent
ESM-1b
MutationAssessor
PHACT
SIFT4G
DeMaSk
PROVEAN
Envision
Tranception
SIFT
sequence_unet
phyloP
BLOSUM62
LRT
GERP++
Grantham
fitCons
SiPhy
phastCons
0
20
40
60
80
100
DMS win rate (%)
Clinical win rate (%)

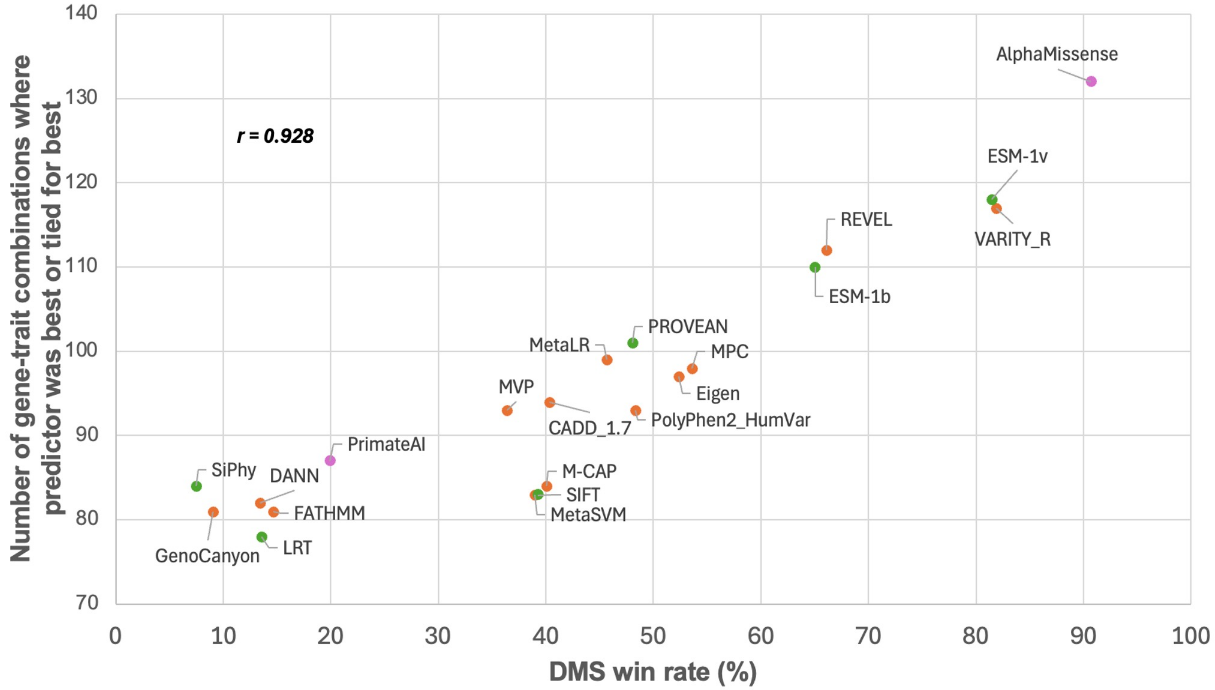


Figure S6. Correlation between performance against the DMS benchmark from this study and performance in inferring human traits from UK Biobank data from Tabet *et al.* The number of gene-trait combinations where each predictor was best or tied for best with the top predictor was taken from panel A of Figure 3 in https://doi.org/10.1186/s13059-024-03314-7.

Research | Open access | Published: 01 July 2024

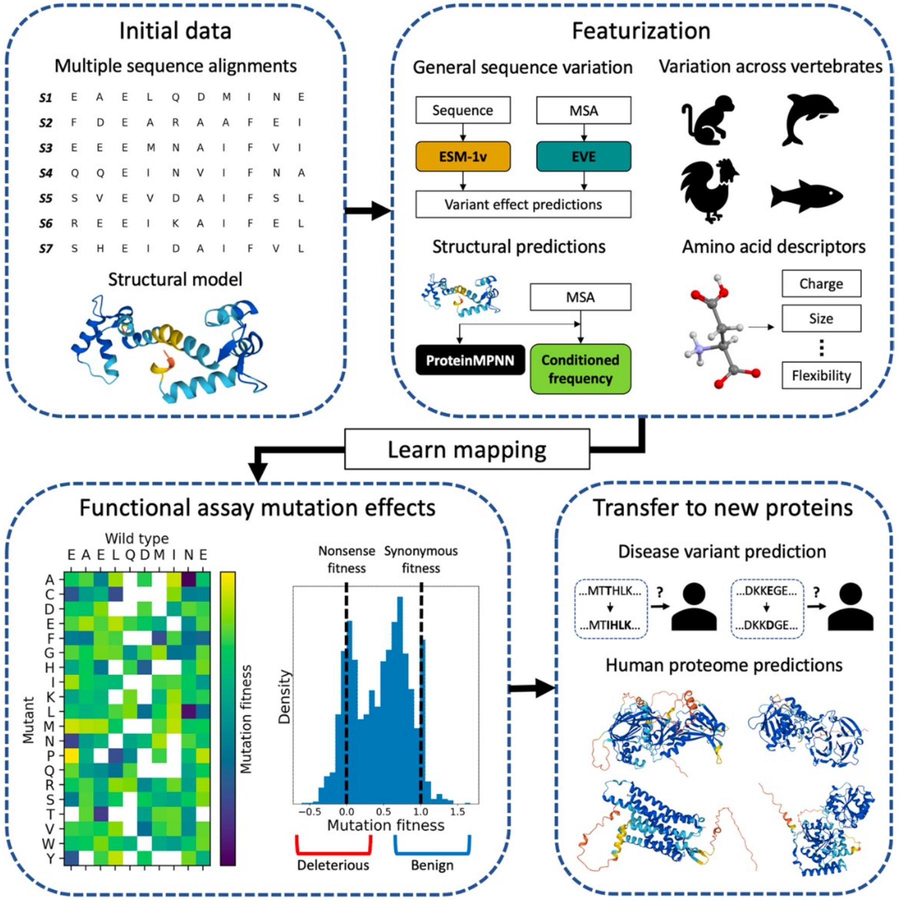
Benchmarking computational variant effect predictors by their ability to infer human traits

Daniel R. Tabet, Da Kuang, Megan C. Lancaster, Roujia Li, Karen Liu, Jochen Weile, Atina G. Coté, Yingzhou Wu, Robert A. Hegele, Dan M. Roden & Frederick P. Roth

Genome Biology **25**, Article number: 172 (2024) | Cite this article

CPT-1 (2023): a DMS-trained model that generalises across protei

- CPT-1 is a population-free cross-protein transfer model trained on deep mutational scanning from five human proteins **(not used in its benchmarking!)**.
- Combines several orthogonal signals:
  - DMS measurements
  - Broad multiple sequence alignments
  - Vertebrate-focused alignments
  - Lightweight structure descriptors
  - Other pop-free models (EVE, ESM-1v) as inputs



Jagota et al. Genome Biology (2023) 24:182
https://doi.org/10.1186/s13059-023-03024-6

Genome Biology

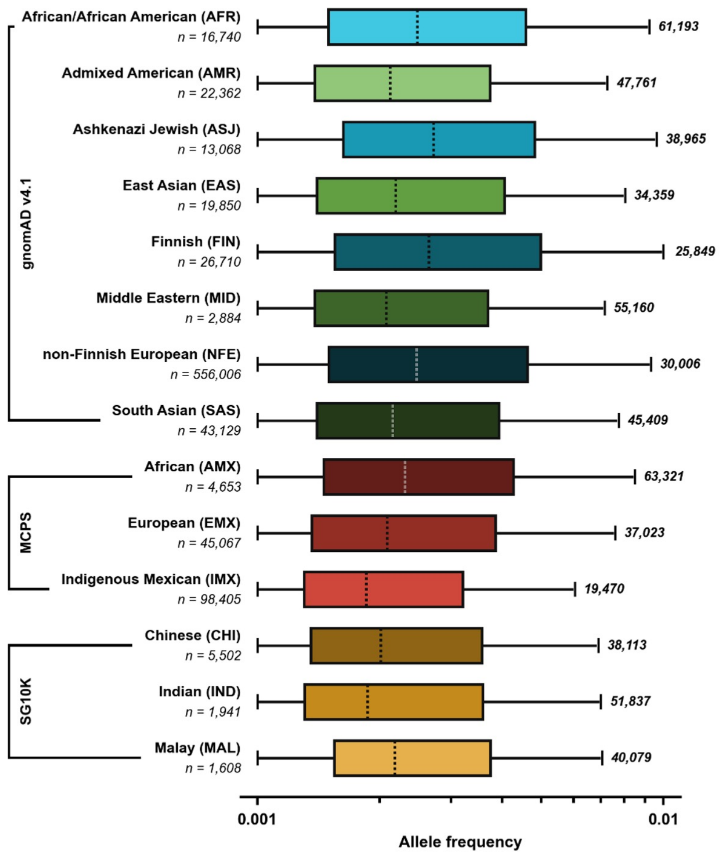
RESEARCH Open Access

Cross-protein transfer learning substantially improves disease variant prediction

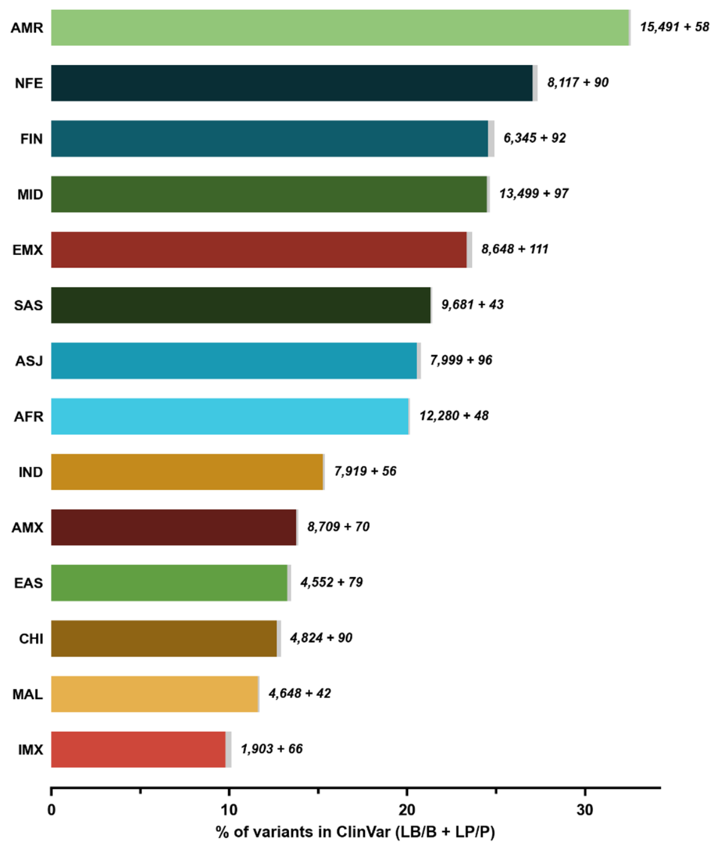
Milind Jagota[1†], Chengzhong Ye[2†], Carlos Albors[1], Ruchir Rastogi[1], Antoine Koehl[2], Nilah Ioannidis[1,3,4] and Yun S. Song[1,2,4*]

Do VEPs show ancestry bias based on their training?

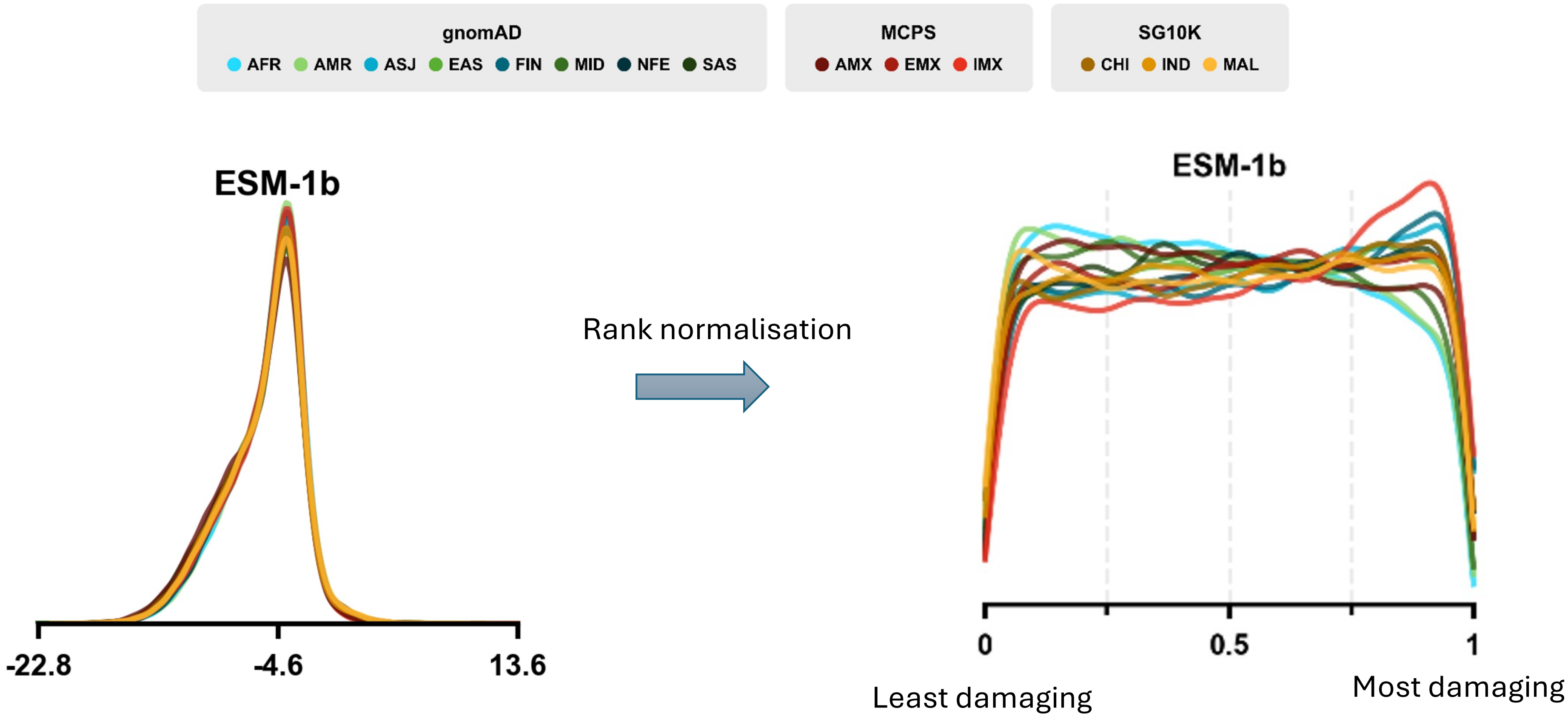
Missense variants observed across 14 populations

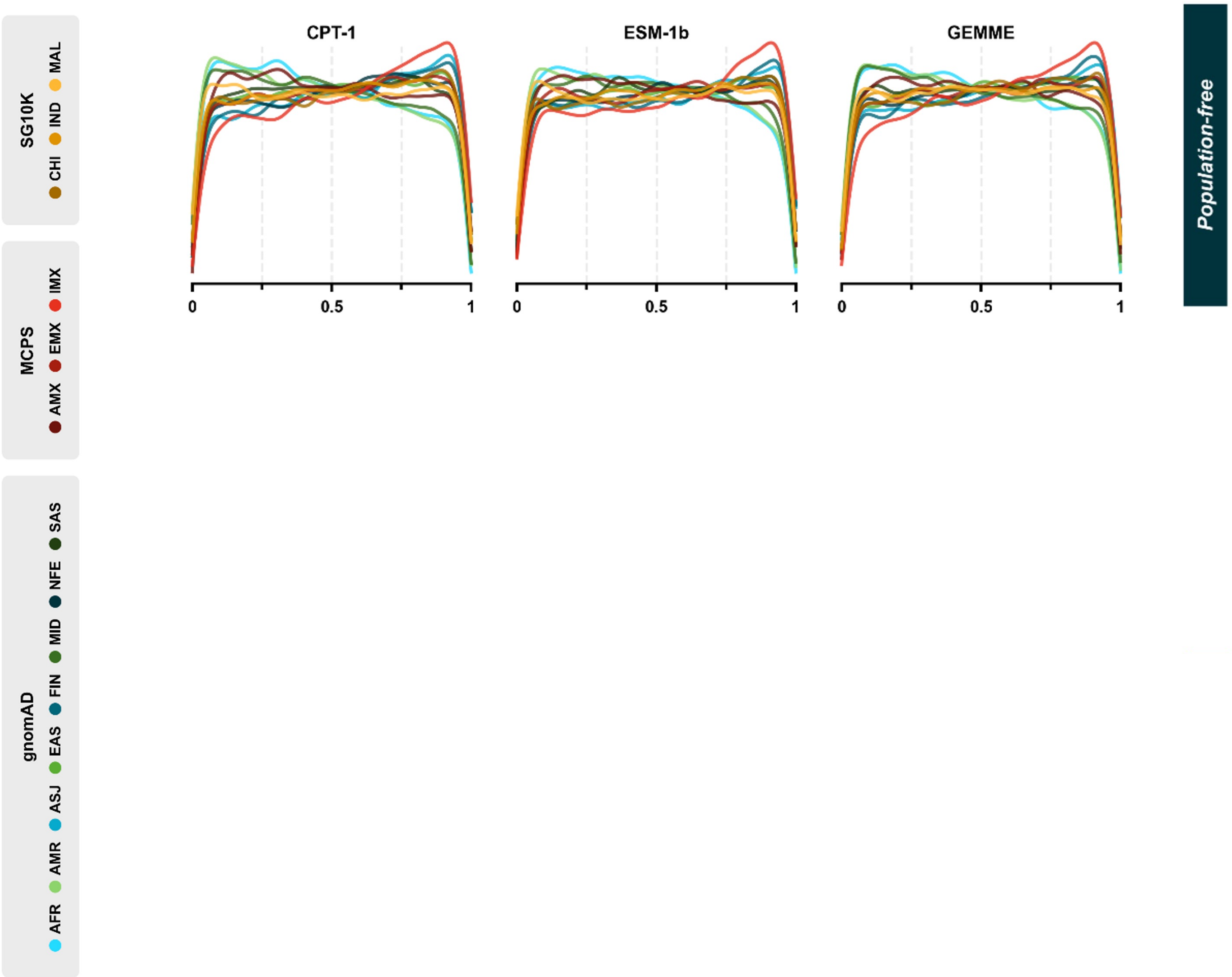


Ascertainment bias in clinical classifications

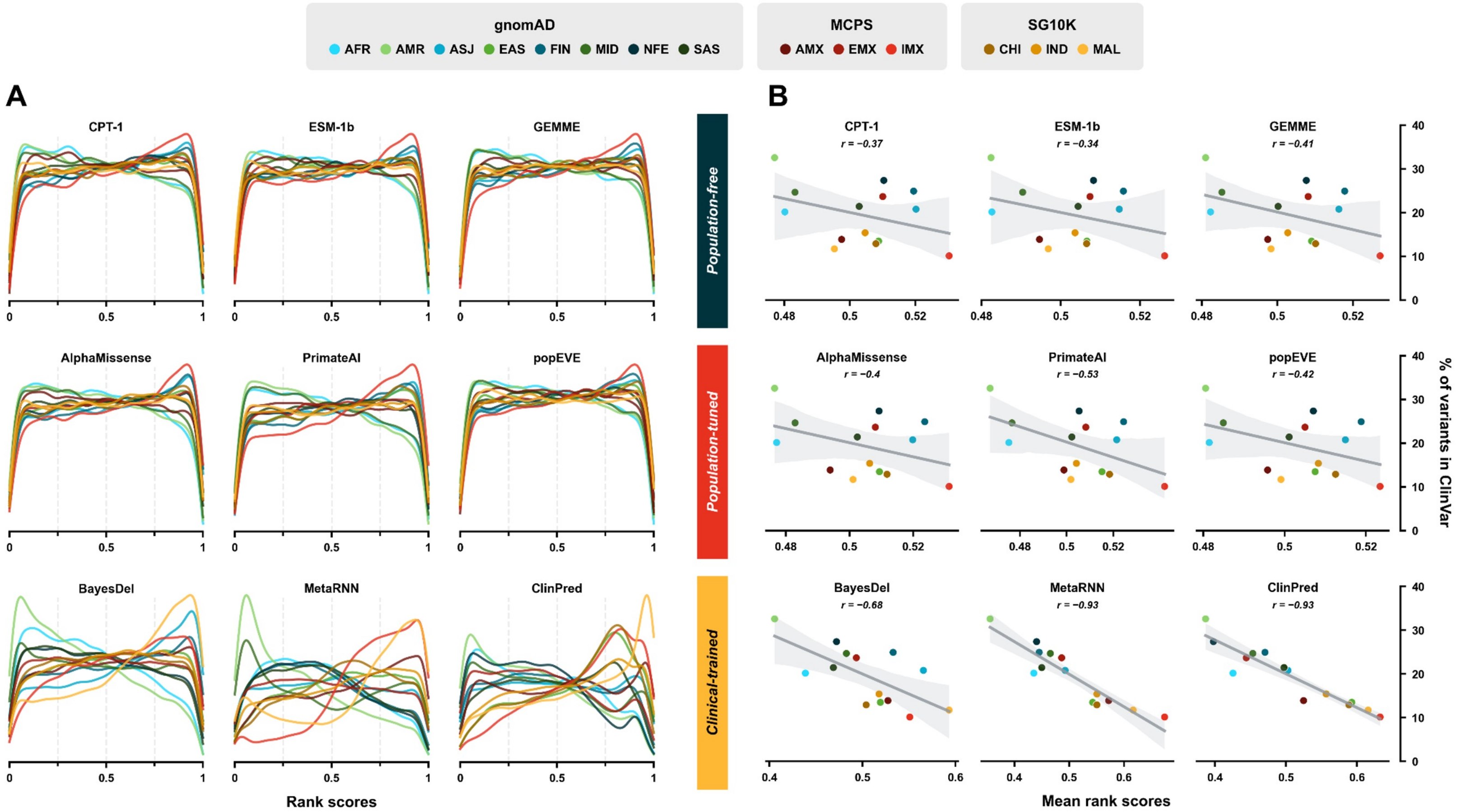


Comparison of VEP score distributions

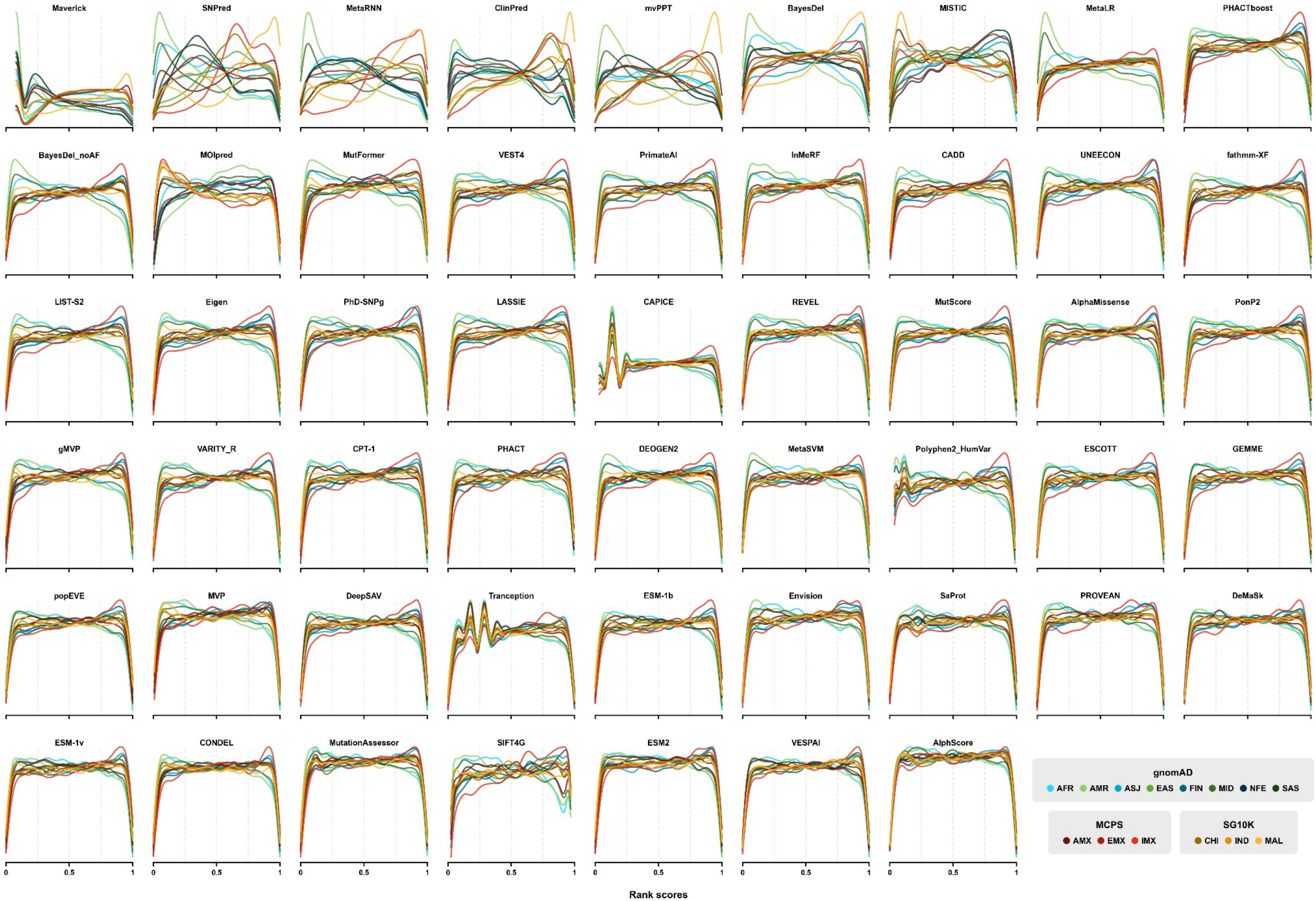



CPT-1
ESM-1b
GEMME
0
0.5
1
Population-free
SG10K
CHI
IND
MAL
MCPS
AMX
EMX
IMX
gnomAD
AFR
AMR
ASJ
EAS
FIN
MID
NFE
SAS

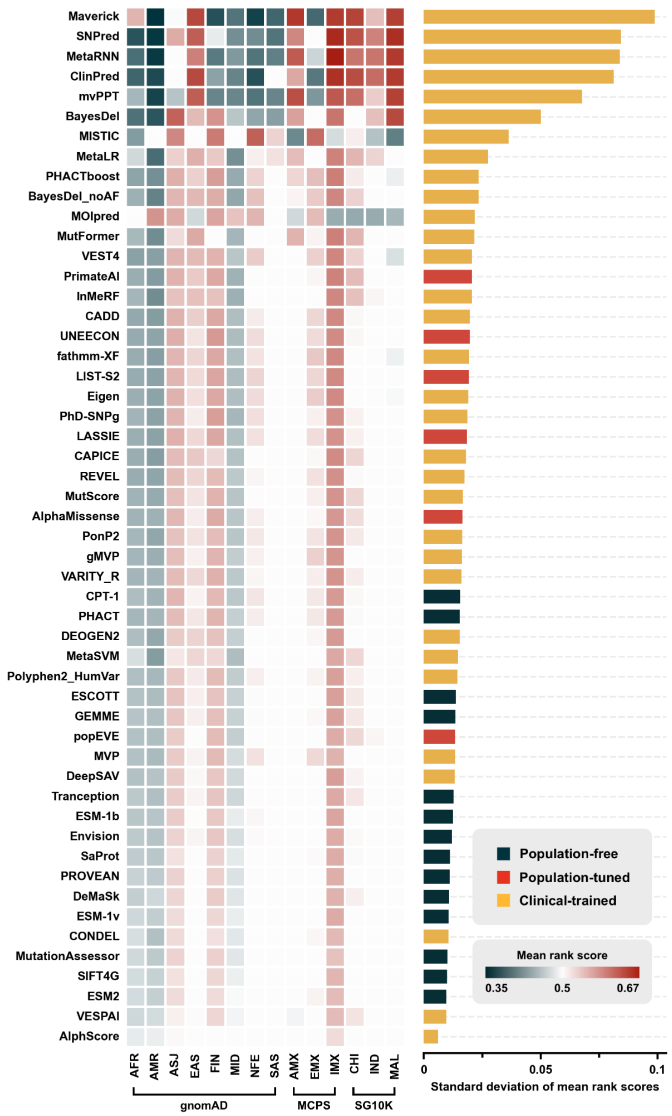
Ascertainment bias appears to explain differences across populations



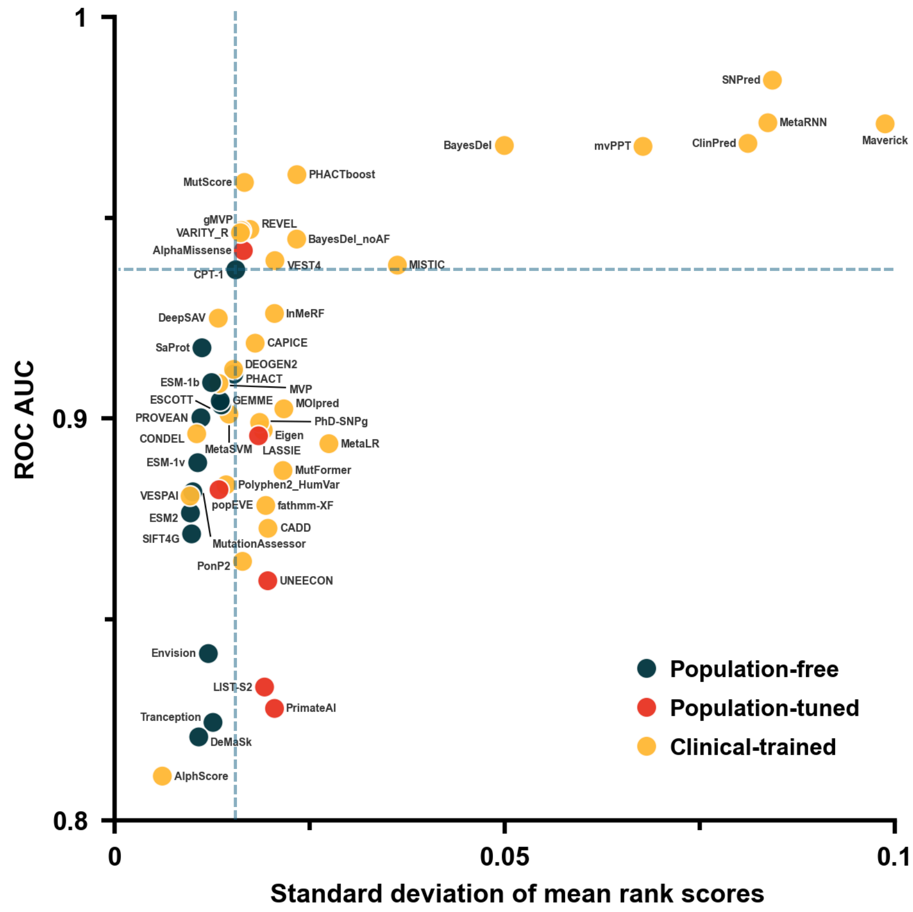
52 VEPs across 14 ancestry groups



52 VEPs across 14 ancestry groups



Pathogenicity prediction vs ancestry bias



How to interpret VEP scores

- VEPs use different score scales and conventions, so values cannot be compared directly across methods.
- Most predictors use higher = more damaging, while others use lower = more damaging (for example SIFT).
- Some outputs represent probabilities, others are likelihood ratios, log-likelihoods, fitness estimates or rank scores.
- Terminology can be misleading. For example, AlphaMissense labels variants as “likely pathogenic” or “likely benign”, but these are model-derived categories and do not correspond to clinical ACMG/AMP definitions.
- Correct interpretation requires knowing what the number represents, what direction it moves in and how the model was calibrated.

The ACMG/AMP Guidelines (2015) and VEPs

- Multiple evidence categories (population, computational/in silico, functional, segregation, de-novo, etc.) each assigned a strength (very strong / strong / moderate / supporting) for or against pathogenicity.
- The "in silico" category covers computational predictions (e.g., amino-acid substitution effect) and is generally assigned as *supporting* evidence only (code PP3 for pathogenic support, BP4 for benign support).
- ***"If all of the in silico programs tested agree on the prediction, then this evidence can be counted as supporting. If in silico predictions disagree, however, then this evidence should not be used in classifying a variant."***
- Vaguely defined, and rarely directly influences variant classifications.

	Benign		Pathogenic			
	Strong	**Supporting**	**Supporting**	**Moderate**	**Strong**	**Very strong**
Population data	MAF is too high for disorder BA1/BS1 **OR** observation in controls inconsistent with disease penetrance BS2			Absent in population databases PM2	Prevalence in affecteds statistically increased over controls PS4	
Computational and predictive data		Multiple lines of computational evidence suggest no impact on gene /gene product BP4 Missense in gene where only truncating cause disease BP1 Silent variant with non predicted splice impact BP7 In-frame indels in repeat w/out known function BP3	Multiple lines of computational evidence support a deleterious effect on the gene /gene product PP3	Novel missense change at an amino acid residue where a different pathogenic missense change has been seen before PM5 Protein length changing variant PM4	Same amino acid change as an established pathogenic variant PS1	Predicted null variant in a gene where LOF is a known mechanism of disease PVS1
Functional data	Well-established functional studies show no deleterious effect BS3		Missense in gene with low rate of benign missense variants and path. missenses common PP2	Mutational hot spot or well-studied functional domain without benign variation PM1	Well-established functional studies show a deleterious effect PS3	
Segregation data	Nonsegregation with disease BS4		Cosegregation with disease in multiple affected family members PP1	Increased segregation data		
De novo data				De novo (without paternity & maternity confirmed) PM6	De novo (paternity and maternity confirmed) PS2	
Allelic data		Observed in *trans* with a dominant variant BP2 Observed in *cis* with a pathogenic variant BP2		For recessive disorders, detected in trans with a pathogenic variant PM3		
Other database		Reputable source w/out shared data = benign BP6	Reputable source = pathogenic PP5			
Other data		Found in case with an alternate cause BP5	Patient's phenotype or FH highly specific for gene PP4			

Richards et al (2015) Genetics in Medicine

Strengthening the PP3/BP4 rules: quantitative calibration of VEPs

- **Problem with the 2015 ACMG/AMP guidelines**
  - In silico tools (PP3/BP4) could only count as *supporting* evidence, and only when *all* tools agreed.
  - No quantitative justification for thresholds or “agreement”.
  - Many predictors were never designed for clinical classification.
- **Pejaver et al. (2022): turning VEP scores into calibrated evidence**
  - Introduces a **probabilistic, Bayesian** framework linking VEP scores to **likelihood ratios** consistent with ACMG/AMP strengths (supporting, moderate, strong, very strong).
  - Uses carefully curated ClinVar and gnomAD sets to avoid circularity; defines **local likelihood ratios** at each score value.
  - Each tool is mapped to **score intervals** corresponding to PP3/BP4 at supporting, moderate, strong, and very strong levels.
- Several modern predictors (e.g., REVEL, MutPred2, VEST4) **meet moderate or strong evidence thresholds** PP3/BP4 to contribute more meaningfully to classification.

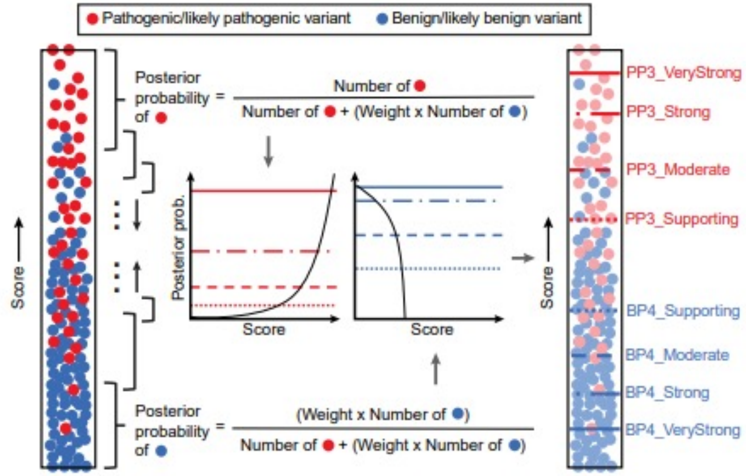
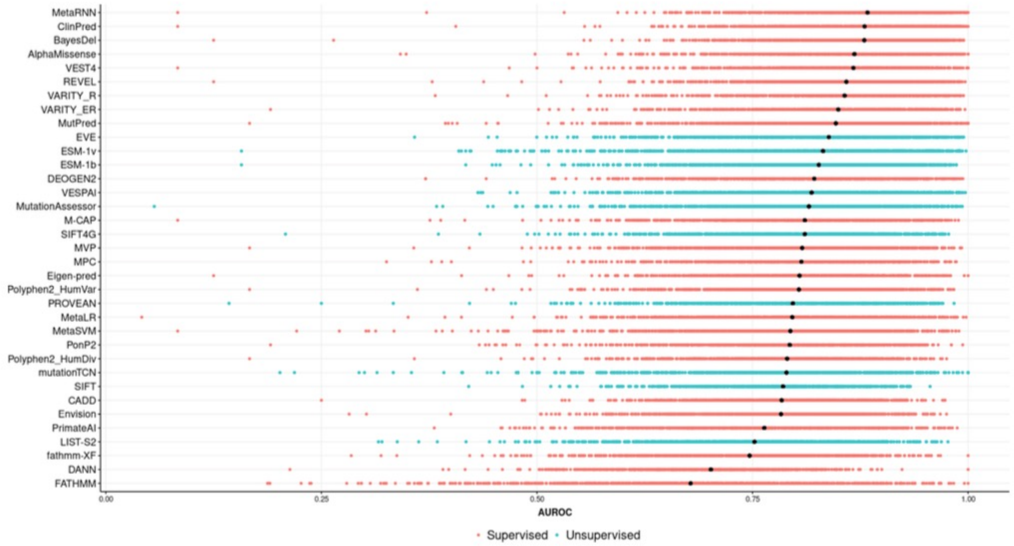


Table 2. Estimated threshold ranges for all tools in this study corresponding to the four pathogenic and four benign intervals

	Benign (BP4)				Pathogenic (PP3)			
Method	Very Strong	Strong	Moderate	Supporting	Supporting	Moderate	Strong	Very Strong
BayesDel	–	–	≤ −0.36	(−0.36, −0.18]	[0.13, 0.27)	[0.27, 0.50)	≥0.50	–
CADD	–	≤0.15	(0.15, 17.3]	(17.3, 22.7]	[25.3, 28.1)	≥28.1	–	–
EA	–	–	≤0.069	(0.069, 0.262]	[0.685, 0.821)	≥0.821	–	–
FATHMM	–	–	≥4.69	[3.32, 4.69)	(−5.04, −4.14]	≤ −5.04	–	–
GERP++	–	–	≤ −4.54	(−4.54, 2.70]	–	–	–	–
MPC	–	–	–	–	[1.360, 1.828)	≥1.828	–	–
MutPred2	–	≤0.010	(0.010, 0.197]	(0.197, 0.391]	[0.737, 0.829)	[0.829, 0.932)	≥0.932	–
PhyloP	–	–	≤0.021	(0.021, 1.879]	[7.367, 9.741)	≥9.741	–	–
PolyPhen2	–	–	≤0.009	(0.009, 0.113]	[0.978, 0.999)	≥0.999	–	–
PrimateAI	–	–	≤0.362	(0.362, 0.483]	[0.790, 0.867)	≥0.867	–	–
REVEL	≤0.003	(0.003, 0.016]	(0.016, 0.183]	(0.183, 0.290]	[0.644, 0.773)	[0.773, 0.932)	≥0.932	–
SIFT	–	–	≥0.327	[0.080, 0.327)	(0, 0.001]	0	–	–
VEST4	–	–	≤0.302	(0.302, 0.449]	[0.764, 0.861)	[0.861, 0.965)	≥0.965	–

Limitations of Pejaver et al

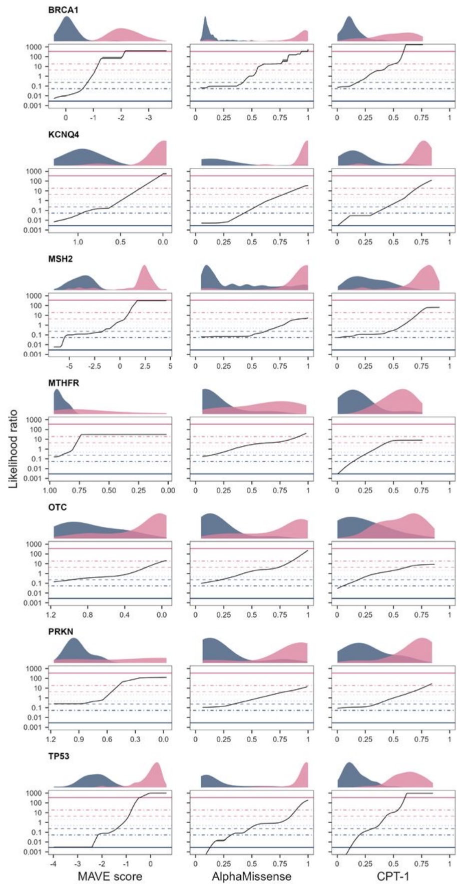
- **Global score thresholds assume global behaviour**, but many VEPs show **large gene-to-gene variability**, with strong performance on some genes and very weak performance on others.
- Calibration using ClinVar variants on clinical-trained VEPs leads to potential circularity issues and risks overstating evidence.



Fawzy and Marsh (2024) Scientific Reports

acmgscaler: Gene-level calibration of VEP (and MAVE) scores

- **Motivation**
  - Predictor behaviour varies widely across genes, so genome-wide thresholds can mislead.
  - *acmgscaler* calibrates **within each gene**, using its own benign and pathogenic variants.
- **What it does**
  - Converts any continuous score into ACMG/AMP evidence strengths via **gene-specific likelihood ratios**.
  - Makes no assumptions about score distributions and works with both VEPs and MAVEs.
- **Key features**
  - Kernel density estimation with bootstrapping to model benign and pathogenic score densities.
  - Regularisation to prevent inflated evidence in sparse regions.
  - Isotonic regression to keep LR curves monotonic.
- **Advantages**
  - Captures **gene-specific heterogeneity** missed by global calibration.
  - Avoids overconfidence when few labelled variants exist.
  - Supports calibration of context-dependent assays and domain-specific effects.
- Implemented as an R package and an easy-to-use Google Colab.



Badonyi & Marsh (2025) Bioinformatics

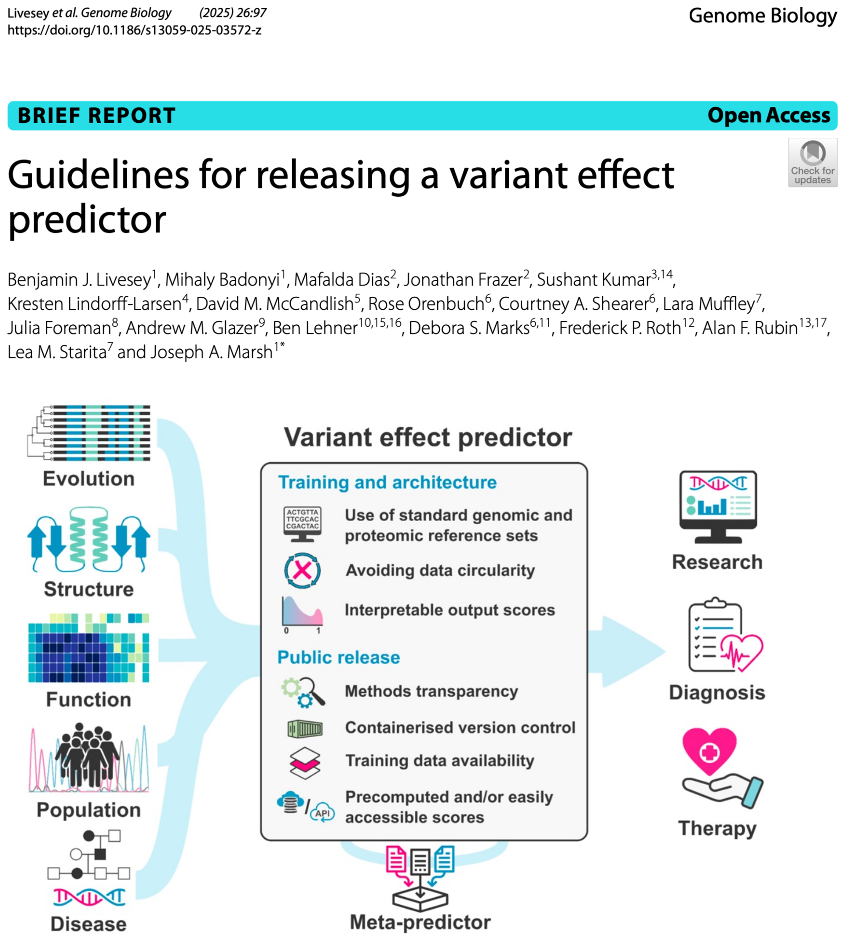

acmgscaler
Benign
Pathogenic
Density resamples
Variant score
N grid points
N resamples
Density resamples interpolated onto a common grid
N grid points
N resamples
Variance-based regularisation of likelihood ratios
Monotonic likelihood ratio estimates
95% confidence interval
Likelihood ratio
Variant score
1
How to run this Colab?
Clicking the ▶ symbol on the left will bring up the upload button*
*Make sure your file is in CSV format (download an example here)
Settings
Name of the column in your data containing the variant labels (at least 10 P and 10 B labels are required)
class_column: class
Name of the column in your data containing the variant scores
value_column: score
Prior probability of pathogenicity* (default 0.1)
*This will not affect the likelihood ratio of pathogenicity, only the ACMG/AMP classification
prior: 0.1
2
Upload your CSV file:
Choose Files No file chosen
Cancel upload
3
1 to 10 of 219 entries
Filter

index	gene	variant	class	score	score_lr	score_lr_lower	score_lr_upper	score_evidence
13	TP53	H115R	B	-3.978137856	0.00315853419735936	0.00301476283970856	0.0033290786130461087	b_strong
63	TP53	S166T	B	-3.25629215	0.00315853419735936	0.00301476283970856	0.0033290786130461087	b_strong
15	TP53	C124S	B	-3.027425108	0.00315853419735936	0.00301476283970856	0.0033290786130461087	b_strong
14	TP53	V122M	B	-2.941846953	0.00315853419735936	0.00301476283970856	0.0033290786130461087	b_strong
87	TP53	S185N	B	-2.883245188	0.00315853419735936	0.00301476283970856	0.0033290786130461087	b_strong
60	TP53	M160I	B	-2.7664013780000003	0.00315853419735936	0.00301476283970856	0.0033290786130461087	b_strong
66	TP53	T170M	B	-2.752953014	0.00315853419735936	0.00301476283970856	0.0033290786130461087	b_strong
86	TP53	S183P	B	-2.747340261	0.00315853419735936	0.00301476283970856	0.0033290786130461087	b_strong
23	TP53	A129D	B	-2.7300762510000003	0.00315853419735936	0.00301476283970856	0.0033290786130461087	b_strong
108	TP53	N210S	B	-2.6487604	0.00315853419735936	0.00301476283970856	0.0033290786130461087	b_strong

Show 10 per page
1 2 10 20 22
4
Plot results (optional)
Show code
Prior: 0.1
p_very_strong
p_strong
p_moderate
p_supporting
b_supporting
b_moderate
b_strong
b_very_strong
Likelihood ratio
Input score

Accessibility of VEPs is inconsistent

- VEPs differ widely in how their data and models are shared.
    - Some provide full proteome score tables, others offer only a web form.
    - Identifier formats vary across tools, including genomic coordinates, transcript-based HGVS and UniProt.
    - File formats are inconsistent, and some tools provide no downloadable data at all.
    - Release channels differ across papers, GitHub, personal websites and unstable servers.
    - Versioning is often unclear, and links are prone to disappearing or changing without notice.
- Why this matters
    - It complicates direct comparison between predictors.
    - It slows down integration into clinical annotation pipelines.
    - It makes unbiased benchmarking much harder than it should be.
    - Researchers spend unnecessary time locating, reformatting and rescuing score files.


Livesey et al. Genome Biology (2025) 26:97
https://doi.org/10.1186/s13059-025-03572-z
Genome Biology

BRIEF REPORT Open Access

Guidelines for releasing a variant effect predictor

Benjamin J. Livesey[1], Mihaly Badonyi[1], Mafalda Dias[2], Jonathan Frazer[2], Sushant Kumar[3,14], Kresten Lindorff-Larsen[4], David M. McCandlish[5], Rose Orenbuch[6], Courtney A. Shearer[6], Lara Muffley[7], Julia Foreman[8], Andrew M. Glazer[9], Ben Lehner[10,15,16], Debora S. Marks[6,11], Frederick P. Roth[12], Alan F. Rubin[13,17], Lea M. Starita[7] and Joseph A. Marsh[1*]

varianteffect.org/veps/

Atlas of Variant Effects

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Variant Effect Predictors

Computational approaches can leverage variant effect data for functional interpretation

Variant Effect Predictors

Guidelines for Releasing a Variant Effect Predictor now available in Genome Biology

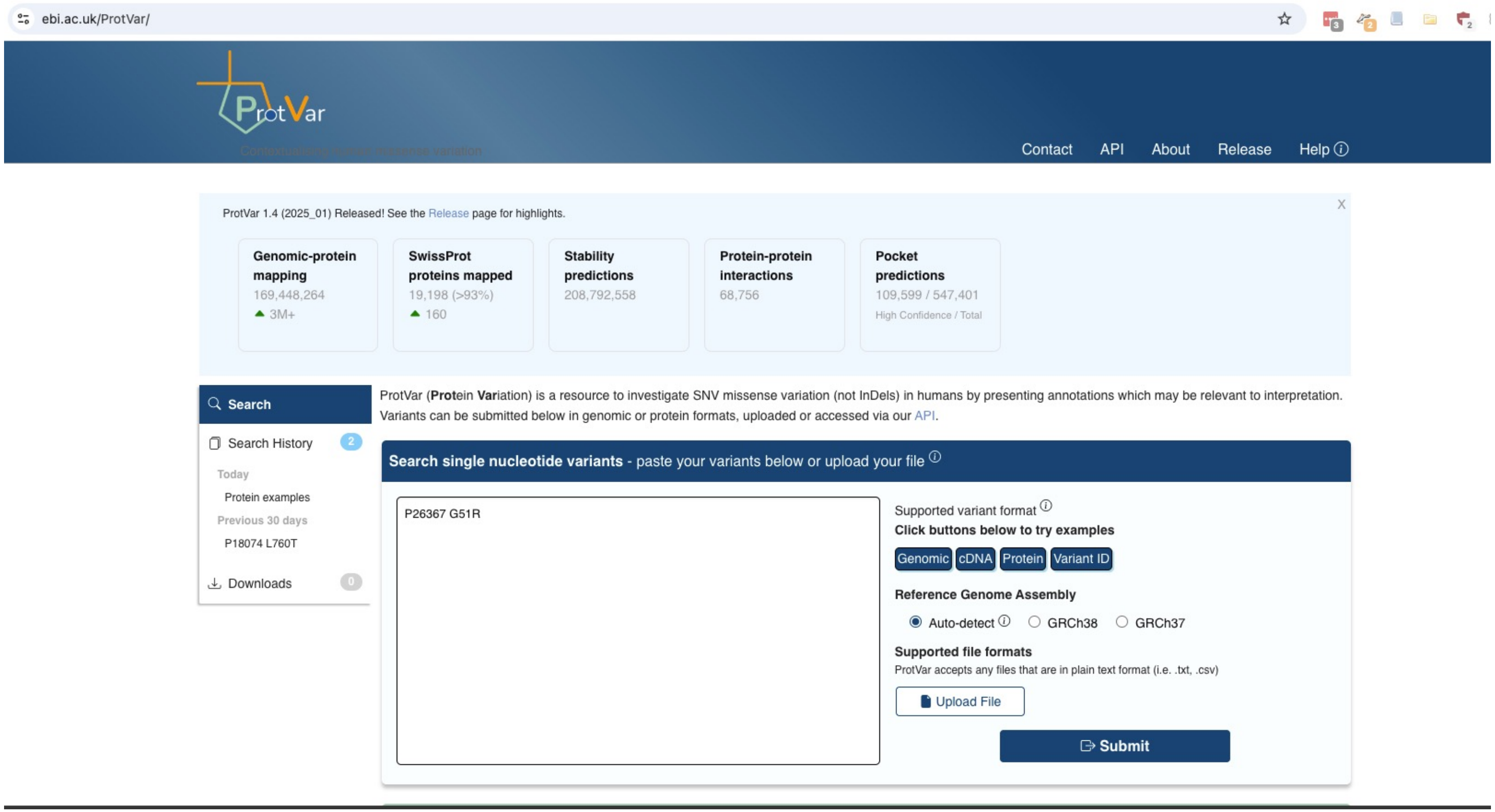
>>Follow this link<< to see our extensive list of variant effect predictors, including classifications, links and references.

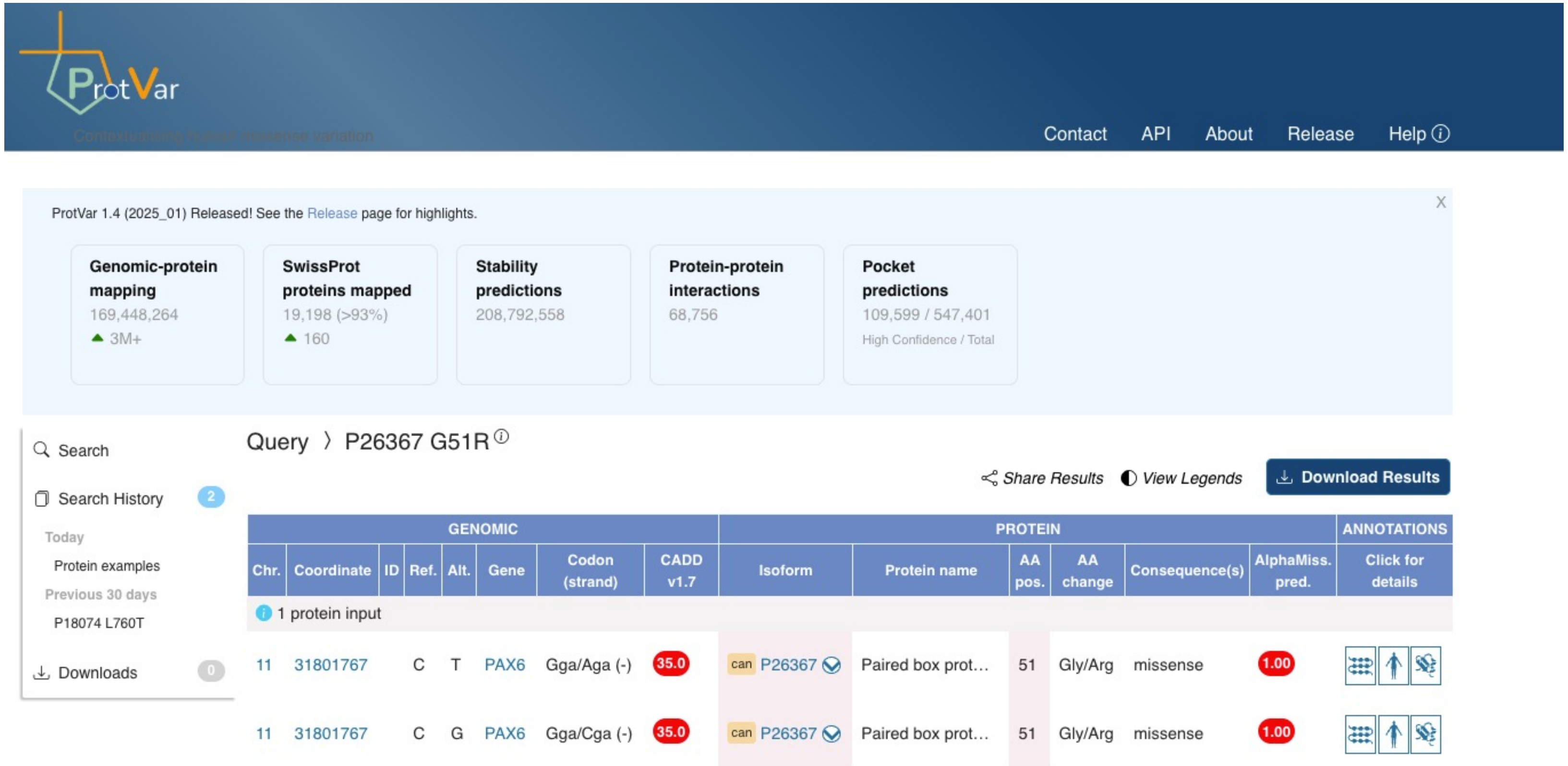
Are we missing something?

If there is a variant effect predictor you think should be added to the list, please tell us at this link: Google Form variantеffect.org/veps

Predictor	Methodology	Type	Feature categories: MSA	Sequence	Function	Structure	Other Predictors	Nucleotide level only?	Author recommended pathogenicity threshold threhold	Link to online predictor	Link to pre-calculated results	Method download
SIFT - Sorting **I**ntolerant **F**rom **T**olerant	Empirical	Population-free	☑	☐	☐	☐	☐	☐	<0.05	https://sift.bii.a-star.edu.sg/	https://sift.bii.a-star.edu.sg/sift4g/ dbNSFP	https://sift.bii.a-star code.html
PolyPhen-2 - **Poly**morphism **Phen**otyping **v2**	Naive Bayes classifier	Clinical-trained	☑	☑	☑	☑	☐	☐	>0.5	http://genetics.bwh.harvard.edu/pph2/	http://genetics.bwh.harvard.edu/pph2/dokuwiki/downloads dbNSFP	http://genetics.bwh pph2/dokuwiki/dow
SNPs&GO	Support vector machine	Clinical-trained	☑	☑	☑	☐	☐	☐	>0.5	https://snps.biofold.org/snps-and-go/snps-and-go.html	Not available	https://hub.docker. snps-and-go
PROVEAN - PROtein **V**ariant **E**ffect **AN**alyzer	Empirical	Population-free	☑	☐	☐	☐	☐	☐	<-2.5	No longer available	https://www.jcvi.org/research/provean#downloads dbNSFP	https://sourceforge provean/
Condel - Consensus **del**eteriousness score	Consensus	Clinical-trained	☐	☐	☐	☐	☑	☐	>0.5	http://bbglab.irbbarcelona.org/fannsdb/	Not available	Not available
MutPred2	Neural network ensemble	Clinical-trained	☑	☑	☐	☑	☐	☐	>0.5	http://mutpred.mutdb.org/	Not available	https://github.com/ pred2
SNPs&GO3D	Support vector machine	Clinical-trained	☑	☑	☑	☑	☐	☐	>0.5	https://snps.biofold.org/snps-and-go/snps-and-go-3d.html	Not available	https://hub.docker. snps-and-go
PON-P2 - **P**athogenic **O**r **N**ot **P**ipeline **v2**	Random forest	Clinical-trained	☑	☑	☑	☑	☐	☐	>0.5	http://structure.bmc.lu.se/PON-P2/	http://structure.bmc.lu.se/PON-P2/ (direct download link on page)	Not available
	Combinatorial								Medium-High >3.5		http://mutationassessor.org/r3/	

ProtVar provides an easy way to get a few individual variant effect scores




ProtVar
Contact
API
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Release
Help
ProtVar 1.4 (2025_01) Released! See the Release page for highlights.
Genomic-protein mapping
169,448,264
3M+
SwissProt proteins mapped
19,198 (>93%)
160
Stability predictions
208,792,558
Protein-protein interactions
68,756
Pocket predictions
109,599 / 547,401
High Confidence / Total
Search
Search History 2
Today
Protein examples
Previous 30 days
P18074 L760T
Downloads 0
Query 〉 P26367 G51R
Share Results
View Legends
Download Results
GENOMIC
PROTEIN
ANNOTATIONS
Chr. Coordinate ID Ref. Alt. Gene Codon (strand) CADD v1.7
Isoform Protein name AA pos. AA change Consequence(s) AlphaMiss. pred.
Click for details
1 protein input
11 31801767 C T PAX6 Gga/Aga (-) 35.0 can P26367 Paired box prot… 51 Gly/Arg missense 1.00
11 31801767 C G PAX6 Gga/Cga (-) 35.0 can P26367 Paired box prot… 51 Gly/Arg missense 1.00

11	31801767	C	T	PAX6	Gga/Aga (-)	35.0	can P26367	Paired box prot…	51	Gly/Arg	missense	1.00

Functional information

Variant Residue Position	Region Containing Variant Position

Annotations from UniProt

No functional data for the variant position

Glycine Arginine

Predictions

Conservation	0.96	very high
Structure predictions		
Stability change ΔΔG	0.06	unlikely to be destabilising
Pathogenicity predictions		
CADD	35.0	highly likely deleterious
AlphaMissense	1.00	pathogenic
EVE	0.98	pathogenic
ESM-1b	-11.5	pathogenic

- [x] ProtVar standardised colours

Annotations from UniProt

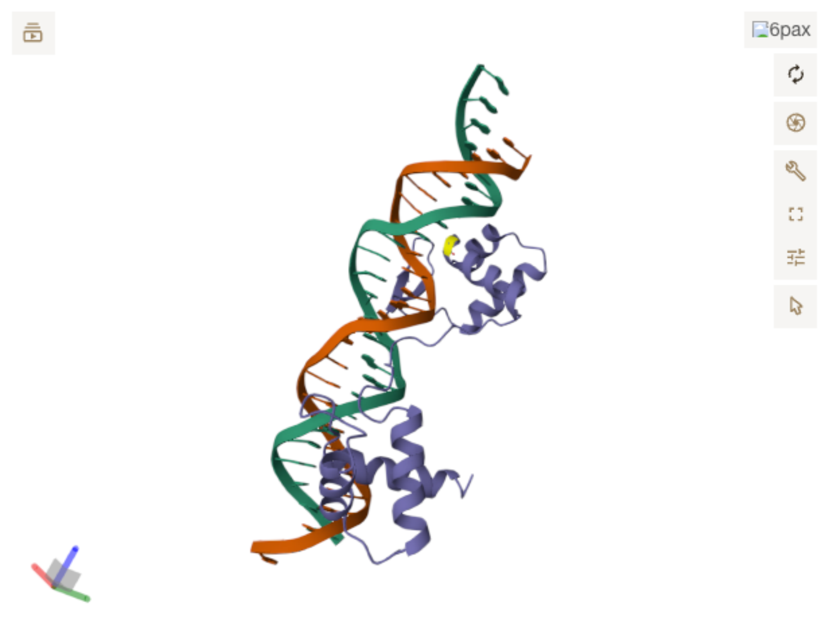
- Chain-Paired box protein Pax-6
- DNA Binding Residue-Paired
- Region-PAI subdomain
- Alpha-helix

Structure predictions

- Pockets containing variant
- Protein-protein interfaces containing variant

11	31801767	C	T	PAX6	Gga/Aga (-)	35.0	can P26367	Paired box prot…	51	Gly/Arg	missense	1.00

Structures



Click variant to see surrounding residues
Click white space to zoom out to whole structure

PDBe Experimental Structure				
PDB ID	Chain	PDB pos.	Resolution (Å)	Method
6pax	C	48	2.5	X-ray diffraction

Zoom to variant | Highlight chain C | Reset

Predicted Structure based on AlphaFold		
ID	Position	Pockets
AF-P26367-F1	51	P2
AlphaFill-P26367	51	P2

Predicted Interacting Structure		
Chain A	Chain B	pDockQ
O75603	P26367	0.535
P26367	Q8IW40	0.452
P26367	Q99469	0.353

Practical activity

https://sites.google.com/view/mave2025-lectures

1. Benchmarking VEP exercise
2. VEP score calibration exercise
3. Investigate some variants of uncertain significance (VUS)